PERMUTATIONS & COMBINATIONS

PERMUTATIONS

SYNOPSIS

Fundamental Principle :

➢ **Multiplication Principle:** If an operation can be performed in 'm' different ways and another operation in 'n' different ways then these two operations can be performed one after the other in 'mn' ways.

Note: Here the key word is **and,** so **and** represents multiple rule

W.E-1: ***In a shelf there are '2' different physics books and '3' different chemistry books. The number of ways in which a student can select a physics book and chemistry book is***

Sol. A student can select a physics book and a chemsitry book in $2 \times 3 = 6$ ways as explained below. Let the physics books be P_1, P_2 and chemistry books be C_1, C_2, C_3.

The six selections are P_1C_1, P_1C_2, P_1C_3, P_2C_1, P_2C_2, P_2C_3.

W.E-2: ***A two wheeler agent sells scooters, motorcycles. In each body pattern two capacities 100 c.c. and 150 c.c. available. In each capacity there are five colours. The number of choices a customer will have to buy a vehicle is***

Sol. The number of choices a customer will have to buy a vehicle is 2 x 2 x 5 = 20.

➢ **Addition Principle:** If an operation can be performed in 'm' different ways or another operation in 'n' different ways then either of these two operations can be performed in 'm + n' ways (provided only one has to be done).

Note 1: Here key word is **or,** so **or** represents Addition rule

Note 2: This principle can be extended to any finite number of operations.

W.E-3: ***In a shelf there are '2' different physics books and '3' different chemistry books. The number of ways in which a student can select a physics book or a chemistry book is***

Sol. A student can select a physics book or a chemistry book in (2 + 3) = 5 ways.

The five ways are P_1, P_2, C_1, C_2, C_3.

W.E-4: ***There are 15 two bed room flats in a building and 10 two bed room flats in other building and 8 two bed room flats in a third building. The number of choices a customer will have for buying a flat is***

Sol. The number of choices a customer will have for buying a flat is 15 + 10 + 8 = 33.

Factorial 'n'

➢ The continuous product of first 'n' natural numbers is called factorial n and is denoted by n! (or) $\angle n$ i.e., n! = 1 x 2 x 3 x .... (n-1) x n.

n! = n [(n-1)!]

➢ 0! = 1 | 6! = 720

1! = 1 | 7! = 5040

2! = 2 | 8! = 40320

3! = 6 | 9! = 362880

4! = 24 | 10! = 3628800

5! = 120

$(2n)! = 2^n . n! . [1.3.5 ... (2n-1)]$

➢ **Permutation:** An arrangement that can be formed by taking some or all of a finite set of things (or objects) is called a **Permutation**.

Order of the things is very important in case of permutation.

➢ **Linear Permutation:** A permutation is said to be a **Linear Permutation** if the objects are arranged in a line. A linear permutation is simply called as a permutation.

W.E.5:- ***From the letters of the word 'RULE'. The number of two letter permutations is***

Sol. From the letters of the word 'RULE', two letter permutations (linear permutations) are RU, UR, RL, LR, RE, ER, UL, LU, UE, EU, LE, EL.

Permutations of 'n' Dissimilar things:

- The number of (linear) permutations that can be formed by taking r things at a time from a set of n dissimilar things (r < n) is denoted by ${}^nP_r$ or P(n, r) or $p\binom{n}{r}$
- The number of permutations of n dissimilar things taken r at a time is equal to the number of ways of filling of r blank places in a row by n dissimilar things.
- ${}^nP_r = n(n-1)(n-2) \dots (n-r+1) = \dfrac{n!}{(n-r)!} = \dfrac{\lfloor n}{\lfloor n-r}$
- The number of permutations of n dissimilar things taken all at a time is ${}^nP_n = n! = \lfloor n$

Properties of ${}^np_r$

- ${}^nP_0 = 1$
- ${}^nP_r = {}^{(n-1)}P_r + r \cdot {}^{(n-1)}P_{r-1}$ or ${}^nP_r + r \cdot {}^nP_{r-1} = {}^{(n+1)}P_r$
- ${}^nP_r = n\,{}^{(n-1)}P_{(r-1)} = n(n-1) \cdot {}^{(n-2)}P_{(r-2)}$ etc.
- $\dfrac{{}^nP_r}{{}^nP_{r-1}} = (n-r+1)$ and $\dfrac{{}^nP_r}{{}^{(n-1)}P_{r-1}} = n$
- If r < s < n, then ${}^nP_s$ is divisible by ${}^nP_r$.
- If $\dfrac{{}^nP_{r-1}}{a} = \dfrac{{}^nP_r}{b} = \dfrac{{}^nP_{r+1}}{c}$ then $b^2 = a(b+c)$
- $1.{}^1P_1 + 2.{}^2P_2 + \dots + n.{}^nP_n = {}^{(n+1)}P_{n+1} - 1$
- Number of permutations of n different things, taken r at a time, when a particular thing is to be always included in each arrangement, is $r.{}^{n-1}P_{r-1}$
- Number of permutations of n different things, taken r at a time, when a particular thing is never taken in each arrangement is ${}^{n-1}P_r$

W.E-6: ***The number of permutations of 6 different things taken 4 at a time, when a particular thing is to be always included in each arrangement is***

Sol. A particular thing is to be always included in each arrangement is $4 \times {}^5P_3$

W.E-7: ***The number of permutations of 6 different things taken 4 at a time, when a particular thing is never taken in each arrangement is***

Sol. A particular thing is never taken in each arrangement is ${}^5P_4$.

- Number of permutations of n different things, taken all at a time, when m specified things always come together is m!.(n - m + 1) !
- Number of permutations of n different things, taken all a time when m specified things never come together is n! - [m! . (n - m + 1)!]
- The number of permutations of 'n' dissimilar things taken 'r' at a time when 'k' particular things never occur is ${}^{(n-k)}P_r$.
- The number of permutations of 'n' dissimilar things taken 'r' at a time when k (<r) particular things always occur is ${}^{(n-k)}P_{(r-k)} \cdot {}^rP_k$.
- The number of ways in which n (first type of different) things and (n-1) (second type of different) things can be aranged in a row so that no two things of same type come together is n! (n-1)!.
- The number of ways in which n (first type of different) things and n (second type of different) things can be arranged in a row alternatively is 2 . n! . n!.

W.E-8: ***The number of ways of arranging 8 boys and 8 girls in a row in which boys and girls come alternately is***

Sol. The number of ways of arranging 8 boys and 8 girls in a row in which boys and girls come alternately is 2 x 8! x 8!.

- To arrange boys and girls in a row alternately, they should be equal in number or with difference 1. Other wise it is not possible to arrange them alternately in a row.

Divisibility:

- The product of n consecutive natural numbers is divisible by n!
- A number is divisible by 2, if the last digit is even.
- A number is divisible by 3, if the sum of the digits in the number is divisible by 3.

W.E-9: ***The number of 4 - digit numbers that can be formed using the digits 2, 3, 5, 6, 8 (without repetition), which are divisible by 3***

Sol. A number is divisible by 3, if the sum of digits in it is a mulitple of 3. Since the sum of the given 5 digits is 24, we have to leave either 3 or 6 and use the digits 2, 5, 6, 8 or 2, 3, 5, 8. In each case, we can permute them in 4! ways. Thus the number of 4-digit numbers divisible by 3 is
2 x 4! = 48.

- A number is divisible by 4, if the last two digits of the number formed is divisible by 4.
- A number is divisible by 5, if the last digit is either 0 or 5.
- A number is divisible by 6, if the number is divisible by 2 and 3. (Even numbers whose sum of digits is divisible by 3).

W.E-10: ***The number of 4 digit numbers which are divisible by 6 by using the digits 2, 3, 5, 8.***

Sol. Given digits are 2, 3, 5, 8. A number is divisible by 6 means it is divisible by 2 and 3. A number is divisible by 3, if the sum of the digits is multiple of three. Here 2 + 3 + 5 + 8 = 18 = 3 x 6.

A number is divisible by 2, if the units digit is even number

$$\_\_\quad \_\_\quad \_\_\quad \underset{(2\text{ ways})}{\overset{(2,\,8)}{\_\_}}$$

Here unit's place filled in 2 ways (2 or 8).

Now remaining '3' gaps are filled in $\lfloor 3$ ways.

$\therefore$ required numbers which are divisible by 6 are = $\lfloor 3 \times 2 = 12$.

- A number is divisible by 7, if the difference between twice the digit in the units place and the number formed by the other digits is either 0 or a multilple of 7.

W.E-11: ***Is this number 3675 is divisible by 7.***

Sol. Let the number is 3675

Here units place digit is 5

Twice the unit's place = 2 x 5 = 10

Now the remaining number is '367'

Then the difference between 367 and 10 is 357

clearly 357 is divisible by '7'

$\therefore$ 3675 is divisible by '7'.

- A number is divisible by 8, if the number formed by the last three digits is divisible by 8.
  Example : 2192,9128
- A number is divisible by 9, if the sum of its digits is divisible by 9.
  Example : 6453, 8640
- A number is divisible by 10, if the last digit is 0.
- A number is divisible by 11, if the sum of the digits in the odd places and the sum of the digits in the even places are equal or differ by a multiple of 11.
  Example : 209, 3564

Sum of Numbers: Sum of the numbers formed by taking all the given n digits (excluding 0) is (Sum of all the n digits) x (n-1)! x (111 ... n times).

- Sum of the numbers formed by taking all the given n digits (including 0) is (sum of all the n digits)[(n-1)! x (111 .... n times) - (n-2)! (111 .... (n-1) times)]
- Sum of all the r-digit numbers formed by taking the given n digits (excluding 0) is (sum of all the n digits) x ${}^{(n-1)}P_{r-1}$ x (111 ...... r times).
- Sum of all the r-digit numbers formed by taking the given n digits (including 0) is (sum of all the n digits) $[{}^{(n-1)}P_{r-1}$ x (111 .... r times) - ${}^{(n-2)}P_{r-2}$ x (111 ... (r-1) times)].

W.E-12: ***Find the sum of all 4 digited numbers that can be formed using the digits 0, 2, 4, 7, 8 without repetition.***

Sol. Put n = 5, r = 4

$$\Rightarrow (0+2+4+7+8)\times[{}^{5-1}P_{4-1}\times(1111)-{}^{5-2}P_{4-2}(111)]$$

$$= 21[({}^{4}P_{3}\times(1111)-{}^{3}P_{2}(111)]$$

$$= 21[(24\times(1111)-6(111)] = 21[(26664-666)]$$

$$= 21\times 25998 = 545958$$

- The sum of the digits in any place of all the numbers formed with the help of $a_1, a_2, \ldots. a_n$ (excluding zero) taken all at a time is
  $(n-1)!.(a_1+a_2+.......+a_n)$

W.E.13:- ***The sum of the digits at the 100's place of all the numbers formed with the digits of 5,6,7,8 taken all at a time is***

Sol. The sum of the digits at the 100's place of all the numbers formed with the digits of 5, 6, 7, 8 taken all at a time is = (n-1)! x (sum of the given digits) $=(4-1)!\times(5+6+7+8) = 3!\times 26 = 156$

- When n digits are given excluding zero, then the sum of the value of digits in any place of n digited number is (n - 1)! x sum of the numbers x {its place value}.

W.E-14: ***The sum of the 'value' of the digits at the 100's place of all the numbers formed with digits of 5, 6, 7, 8 taken all at a time is***

Sol. Sum of the value of the digits at the 100's place of all the numbers formed with digits of 5, 6,7, 8 taken all at a time is

(n-1)! x (sum of the given digits) x 100

= (4-1)! x (5 + 6 + 7 + 8) x 100

= 3! x 26 x 100 = 6 x 26 x 100 = 15600

Permutations when repetitions are allowed:

- The number of permutations of n dissimilar things taken r at a time when repetition of things is allowed any number of times is n^r.
- The number of permutations of n things (distinct) taken r at a time with atleast one repetition $= n^r - {}^nP_r$.

W.E-15: ***The number of 4 digit numbers that can be formed using the digits 1, 2, 3, 4, 5, 6 that are divisible by 3 when repetition is allowed***

Sol. Fill the first 3 places with the given 6 digits in 6^3 ways. ___ ___ ___ ___ 2 or 8

(6) (6) (6) (2 ways)

Now, after filling up the first 3 places with three digits, if we fill up the units place in 6 ways, we get 6 consecutive positive integers. Out of any six consecutive integers exactly two are divisible by 3. Hence the units place can be filled in 2 ways. Hence the number of 4 digit numbers divisible by 3 is 2 x 6^3 = 432.

W.E-16: ***The number of 4-digit numbers that can be formed using the digits 1, 2, 3, 4, 5, 6 that are divisible by 6 when repetition is allowed.***

Sol. Fill the first 3 places with the given 6 digits in 6^3 ways. ___ ___ ___ ___

(6) (6) (6) (1 way)

Now, after filling up the first 3 places with three digits, if we fill up the units place in 6 ways, we get 6 consecutive positive integers. Out of any six consecutive integers exactly one is divisible by 6. Hence the units place can be filled in one way. Hence the number of 4 digit numbers divisible by 6 is 1 x 6^3 = 216.

W.E-17: ***Let A be a set of n (>3) distinct elements. The number of triplets (x, y, z) of the elements of A in which at least two coordinates are equal is***

Sol. Here at least two coordinates are equal. i.e two (or) three cordinates are equal.

$\therefore$ At least one coordinate is repeated

Hence $n^r - n_{P_r}$ where $r = 3$, $\therefore\ n^3 - n_{P_3}$

- The number of permutations of n different things, taken not more than r at a time, when each thing may occur any number of times

$$= n + n^2 + n^3 + ...+ n^r = \frac{n(n^r - 1)}{n-1}$$

W.E-18: ***The number of positive integers having atmost 20 digits using the digits 1, 2, 3, 4, 5, 6.***

Sol. Required number of numbers

$$= 6 + 6^2 + 6^3 + ...+ 6^{20} = \frac{6(6^{20} - 1)}{5}$$

- The number of permutations of n different things taken not more than 'r' at a time (without repetition) $= {}^nP_1 + {}^nP_2 + {}^nP_3 + ... + {}^nP_r$
- Number of positive numbers not more than P digits, if the digits 0, 1, ...,9 are used when repetitions of digits are allowed is $n^P - 1$ where n is number of digits including zero.

W.E-19: ***The number of positive integers having atmost 10 digits using 0,1,2,3,4,5,6,7***

Sol. Required number of numbers = 8^{10} - 1.

- The number of permutations of 'n' different things taken more than 'r' at a time when repetitions are allowed is $(r \le n)$

$$n^{r+1} + n^{r+2} + ...+ n^n = \frac{n\left(n^n - n^r\right)}{n-1}$$

Number of functions :

- The number of mappings that can be defined from set A containing 'm' elements to set B containing 'n' elements is n^m.
- The number of injections (one one functions) that can be defined from set A containing 'm' elements to set B containing 'n' elements is ${}^nP_m$, where $n \ge m$
- The number of surjections (on to functions) that can be defined from set A containing 'm' elements to set B containing '2' elements is $2^m - 2$
- The number of bijections (one one and onto functions) that can be defined from set 'A' containing 'n' elements to set B containing 'n' elements is $n!$
- The number of many to one functions that can be defined from set A containing 'm' elements to set B containing 'n' elements is $n^m - {}^nP_m$

- Number of constant functions that can be defined from set A containing 'm' elements to set B containing 'n' elements is 'n'.

W.E-20: ***The number of constant mappings from A = {1, 2} to B = {a, b,.....,z} is***

Sol. The number of constant functions is equal to n(B)
$\therefore$ The number of constant functions is 26

- The number of surjections (onto functions) that can be defined from set A containing 'm' elements to set B containing 'n' elements is

$$n^m - {}^nC_1(n-1)^m + {}^nC_2(n-2)^m - {}^nC_3(n-3)^m + ...$$

- The Number of **into** functions that can be defined from set A containing 'm' elements to set B containing 'n' elements is Total number of functions - Number of onto functions

W.E-21: ***The number of into functions that can be defined from A = {x, y, z, w, t} to B = $\{\alpha, \beta, \gamma\}$ is***

Sol: n(A) = m = 5, n(B) = n = 3
The number of into functions = (Total no. of functions) **–** (number of onto functions)

$= n^m - (n^m - {}^nc_1.(n-1)^m + {}^nc_2(n-2)^m - {}^nc_3(n-3)^m + ...)$

$= 3^5 - (3^5 - 3c_1.(2)^5 + 3c_2(1)^5 - 3c_3(0)^5)$

$= 243 - (243 - 3\times 32 + 3\times 1 - 0)$

$= 243 - (243 - 96 + 3) = 243 - (150) = 93$

- Number of identity functions that can be defined from set 'A' to itself is ''one"

Palindrome:

- A number or a word which reads same either from left to right or from right to left is called a Palindrome. Some examples of palindromes are ATTA, ROTOR, 12321, 120021 etc.,

Note: The number of palindromes with 'r' distinct letters that can be formed using given 'n' distinct letters is i) $n^{r/2}$ if r is even ii) $n^{\frac{r+1}{2}}$ if r is odd.

W.E-22: ***The number of Palindromes with 6 digits that can be formed using the digits 1, 3, 5, 7, 9.***

Sol. n = 5, r = 6 (even)
$\therefore$ number of Palindromes = $n^{r/2} = 5^{6/2} = 5^3$

Note: The number of palindromes with 'r' distinct digits that can be formed using given 'n' distinct digits (including '0') is

i) $(n-1)\, n^{\frac{r-2}{2}}$ if r is even ii) $(n-1)\, n^{\frac{r-1}{2}}$ if r is odd.

W.E-23: ***The number of Palindromes with '6' digits that can be formed using the digits 0, 2, 4, 6, 8 is***

Sol: Given digits
0, 2, 4, 6, 8.r = 6 (even), n = 5

$\overline{(4)}\ \overline{(5)}\ \overline{(5)}\ \overline{(1)}\ \overline{(1)}\ \overline{(1)}$

$\therefore$ Required 6 digits Palindromes are

$= (n-1)\, n^{\frac{r-2}{2}} = 4\times 5^2$

Circular Permutations:

- A permutation is said to be a **Circular Permutation** if the objects are arranged in the form of a circle or (a closed curve).
- The number of circular permutations of 'n' dissimilar things taken 'r' at a time (when clock wise and anticlock wise orders are taken as different) is $\frac{{}^np_r}{r}$

W.E-24: ***The number ways to arrange 8 persons around a circle by taking 4 at a time.***

Sol. We know that the number of circular permutations of n distinct things taken r at a time

$= \frac{{}^nP_r}{r} = \frac{{}^8P_4}{4}$

- The number of circular permutations of 'n' dissimilar things taken all at a time is (n-1)!.
- The number of circular permutations of 'n' things taken 'r' at a time in one direction (when clock wise and anticlock wise orders are not different) or (hanging type) is $\frac{{}^np_r}{2r}$.

W.E-25: ***Number of circular permutations of 15 things taken 5 at a time in one direction is***

Sol. Required number of ways $= \frac{{}^nP_r}{2r} = \frac{{}^{15}P_5}{10}$

- The number of circular permutations of 'n' dissimilar things in clock-wise direction = Number of circular permutations in anticlock-wise direction $= \frac{(n-1)!}{2}$

Note: We have to treat garlands of flowers, chains of beads, necklace of beads etc as hanging type of arrangements.

W.E-26: ***The number of ways of preparing a chain with 5 different coloured beads is***

Sol. The number of ways of preparing a chain with 5 different coloured beads $= \frac{(n-1)!}{2}$, (n = 5)

$= \frac{(5-1)!}{2} = \frac{24}{2} = 12$

- The number of ways in which 'n' dissimilar things can be arranged in a circular manner such that no two will have (will not have) same neighbours in any two arrangements is $\frac{(n-1)!}{2}$

Permutations in which some are alike and the rest are different

- The number of permutations of 'n' things taken all at a time when 'p' of them are alike and the rest are different is $\frac{n!}{p!}$
- If 'p' things are alike of one kind, 'q' things are alike of second kind, 'r' things are alike of third kind, then the number of permutations formed with (p + q + r) things is $\frac{(p+q+r)!}{p!\,.\,q!\,.\,r!}$

Specified order or desired order

- The number of permutations of 'n' different things when 'r' things are in specified order is ${}^{n}p_{n-r}\ (or) \frac{n!}{r!}$

W.E-27: ***The number of ways of arranging 15 students $A_1, A_2, A_3, \ldots, A_{15}$ in a row such that A_2 must be seated after A_1 and A_3 must be seated after A_2 is***

Sol. Consider 15 places to arrange 15 persons $A_1, A_2, A_3, \ldots, A_{15}$. Keeping aside A_1, A_2 and A_3, first arrange the remaining 12 persons in any 12 places of these 15 places. It can be done in ${}^{15}P_{12}$ ways. Now we shall be left with 3 places which need not be consecutive. Now arrange A_1, A_2 and A_3 in these 3 places according to the desired order. It can be done in one way only.

$\therefore$ The number of arrangements in which A_1 comes before A_2 and A_2 before A_3 $= {}^{15}P_{12} = \frac{\lfloor 15}{\lfloor 3}$

W.E-28: ***The number of ways in which letters of the word 'DEEPAK' be arranged without changing the order of the vowels is***

Sol. Given word is 'DEEPAK'. The word containing 6 letters. '3' are vowels and remaining are consonants. The number of ways in which the letters of the word 'DEEPAK' be arranged without changing the order of the vowels is ${}^{6}P_3 \times 1 = {}^{6}P_3$.

Rank of a word: If all the arrangements formed with all the letters of the given word are put in alphabetical order then the numerical place of the given word is called it's "Rank".

W.E-29: ***Find the rank of the word 'RAMESH'***

A	E	H	M	R	S

Sol. Write the letters in boxes as alphabetical order

Write all factorials begining with one less than number of boxes i.e., 5! 4! 3! 2! 1! 0!

In RAMESH first letter is R. Then multiply first factorial with number of letters before R. In alphabetical order and this process will continue till end.

By adding the above factorials we get the rank.

$\therefore$ The rank of the word 'RAMESH' is

4 x 5! + 0 x 4! + 2 x 3! + 0 x 2! + 1 x 1! + 0! = 494

Dearrangements:

- The number of ways in which exactly 'r' letters can be placed in wrongly addressed envelopes when 'n' letters are put in 'n' addressed envelopes is ${}^{n}P_r \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots\ldots + (-1)^r \frac{1}{r!}\right]$
- The number of ways in which 'n' different letters can be put in their 'n' addressed envelopes so that all the letters are in the wrong envelopes = $n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots\ldots + (-1)^n \frac{1}{n!}\right]$
- 'n' letters are kept in 'n' addressed envelopes. Then the number of ways that exactly one letter will go wrong is 0.
- The number of ways that all the letters will go into correct addressed envelope is 'one' way.

W.E-30: ***The number of ways of '5' letters put in 5 addressed envelopes so that one letter is in right envelope and all the remaining are placed in wrong envelopes is.***

Sol. One letter is in right envelope means remaining '4' letters are placed in wrong envelopes.

$\therefore$ Required no. of ways $= {}^{5}P_{4}\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right]=45$

W.E-31: Vamshi writes letters to his five friends and addresses the corresponding. The number of ways can the letters be placed in the envelopes so that atleast two of them go into the wrong envelopes is

Sol. Required ways=total ways-all correct ways

($\because$ one letter will never go into wrong envelope)

$= 5! - 1 = 119$

C.U.Q

1. There are 3 routes from Guntur to Madras and 4 routes from Madras to Cochin, in how many different ways a person can travel from Guntur to Cochin via Madras is

1) 21 2) 24 3) 12 4) 20

2. The number of permutations of 8 things taken 'r' at a time is 1680. Then r =

1) 4 2) 5 3) 6 4) 7

3. The number of permutations of 'n' dissimilar things taken 'r' at a time, in which a particular thing always occur is

1) ${}^{(n-1)}P_{(r-1)}$ 2) $(r)^{(n-1)}P_{(r-1)}$ 3) $(r)^{(n-1)}P_{r}$ 4) $(r!)^{(n-1)}P_{(r-1)}$

4. $1+1.{}^{6}P_{1}+2.{}^{6}P_{2}+.........+6.{}^{6}P_{6}=$

1) 720 2) 9787 3) 7! 4) 8!

5. The remainder obtained when 1!+2!+......+100! is divided by 240 is

1. 153 2. 154 3. 155 4. 156

6. The number of ways in which the candidates A_1, A_2, ... A_{10} can be ranked if A_1 and A_2 are next to each other is

1) 9! 2! 2) 9! 3) $\frac{10!}{2!}$ 4) $\frac{9!}{4!}$

7. The number of ways in which the candidates $A_1, A_2, ..., A_{10}$ can be ranked if A_1 is always above A_2 is

1) 9! 2! 2) 9! 3) $\frac{10!}{2}$ 4) 10!

8. The number of ways in which 10 candidates $A_1, A_2, A_3, A_4, ..., A_{10}$ can be ranked if A_1 is just above A_2 is

1) 9! 2! 2) 10! 3) 10! 2! 4) 9!

9. The number of words that can be formed from the letters of the word "INTERMEDIATE" in which no two vowels are together is

1) $6!\,.\,{}^{7}P_{6}$ 2) $\frac{6!}{2!}\cdot\frac{{}^{7}p_{6}}{2!\,3!}$ 3) $\frac{6!\,.{}^{7}p_{6}}{2!\,3!}$ 4) $\frac{(7!)\,{}^{7}p_{6}}{2!\,5!}$

10. The number of ways to rearrange the letters of the word CHEESE is

1) 119 2) 240 3) 720 4) 6

11. If the letters of word 'VICTORY' are arranged in the order of a dictionary. Then rank of the word 'VICTORY' is

1) 3731 2) 3732 3) 3733 4) 3720

12. The rank of the word 'SEASON' is

1) 210 2) 220 3) 230 4) 270

13. The number of ways in which 5 boys and 3 girls can sit around a table so that all the girls do not come together is

1) 4020 2) 4120 3) 4220 4) 4320

14. The number of ways that a garland is made with 18 flowers such that the two specified flowers should be side by side in the garland is

1) 15 2) 120 3) 17! 4) 16!

C.U.Q - KEY

1) 3	2) 1	3) 2	4) 2	5) 1	6) 1	7) 3
8) 4	9) 2	10) 1	11) 3	12) 4	13) 4	14) 4

C.U.Q - HINTS

1. $4\times 3=12$

2. $8_{p_r}=1680\Rightarrow r=4$

3. Synopsis $r.{}^{(n-1)}P_{r-1}$

4. Expanding

5. $1!+2!+3!+4!+5!=153$

6. A_1, A_2 are taken as one object and internally arranged in 2!

Remaining 8 objects + one object = 9 objects

7. $A_1, A_2,..., A_{10}$ are arranged in 10! ways in these half of arrangements A_1 always above A_2

8. (10+1-2)!=9!

9. 2I+3E+1A=6

Step-I = 2I + 3E + 1A = 6

Step-II = NTRMDT = 6

Step-III Six letters arranged in $\frac{6!}{2!}$ ways of these 7 gaps 6 letters of step I are arranged in $\frac{^7P_6}{2!\,3!}$ ways

10. $\frac{6!}{3!}-1$

11. $5\times6!+1\times5!+2\times3!+1=3733$

12. $2\times5!+1\times4!+2\times2!+1\times1!+1=270$

13. 7! - 5!. 3!

14. $\frac{16!}{2!}\times2!$

LEVEL-I (C.W)

FUNDAMENTAL PRINCIPLE

1. A student has 5 pants and 8 shirts. The number of ways in which he can wear the dress in different combinations is

1) $^8P_5$ 2) $^8C_5$ 3) 8! x 5! 4) 40

2. An automobile dealer provides motor cycles and scooters in three body patterns and 4 different colours each. The number of choices open to a customer is

1) $^5C_3$ 2) $^4C_3$ 3) 4 x 3 4) 4x3x2

3. In a class there are 10 boys and 8 girls. The teacher wants to select either a boy or a girl to represent the class in a function. The number of ways the teacher can make this selection.

1) 18 2) 80 3) $^{10}P_8$ 4) $^{10}C_8$

PROBLEMS BASED ON ^nP_r FORMULA

4. If $^9P_5+5.\,^9P_4={}^{10}P_r$ then r =

1) 4 2) 5 3) 6 4) 7

5. $\frac{^{20}P_{r-1}}{a}=\frac{^{20}P_r}{3}=\frac{^{20}P_{r+1}}{4}$ **then a =**

1) 9/7 2) 7/9 3) 3/4 4) 4/3

6. If $^{(2n+1)}P_{n-1}:{}^{(2n-1)}P_n=3:5$ then n =

1) 4 2) 5 3) 6 4) 3

7. If $^nP_7=42\,^nP_5$ then n = **[EAM-98]**

1) 5 2) -1 3) 7 4) 12

8. If $^{15}P_r=32760$, then r =

1) 4 2) 5 3) 6 4) 7

9. If $^nP_r=5040$ then (n, r) =

1) (9, 4) 2) (10, 4) 3) (11, 3) 4) (11, 4)

10. The value of $\sum_{r=1}^{10} r.P(r,r)$ is

1) P(11, 11) 2) P(11, 11) - 1

3) P(11, 11) + 1 4) $\underline{|12}-1$

PROBLEMS BASED ON REMAINDER & OTHER MODELS

11. The value of 1 + 1.1! + 2.2! + 3.3! + ... + n.n! is

1) (n+1)!+1 2) (n-1)!+1 3) (n+1)!-1 4) (n+1)!

12. The last two digits in $X=\sum_{k=1}^{100} k!$ are

1) 10 2) 11 3) 12 4) 13

13. The remainder obtained when 1!+2!+......49! is divided by 20 is

1. 13 2. 33 3. 12 4. 11

14. When n! + 1 is divided by any natural number between 2 and n then remainder obtained is

1) 1 2) 2 3) 3 4) 4

15. If $\sum_{k=1}^{m}\left(k^2+1\right)k!=1999(2000!)$, then m is

1) 1999 2) 2000 3) 2001 4) 2002

PERMUTATIONS OF DISSIMILAR THINGS

16. The number of different signals can be given by using any number of flags from 4 flags of different colours is

1) 24 2) 256 3) 64 4) 60

17. Three Men have 4 coats 5 waist Coats, and 6 caps. The number of ways they can wear them is

1) $^{15}P_3$ 2) $4^3 5^3 6^3$ 3) $^4P_3\,^5P_3\,^6P_3$ 4) 180

18. A railway carriage can seat 5 each side. The number of ways a party of 4 girls and 6 boys can seat themselves so that the girls may always have the corner seats is

1) 17,430 2) 17,431 3) 17,280 4) 17,281

19. The number of words that can be formed using any number of letters of the word "KANPUR" is

1) 720 2) 1956 3) 360 4) 370

20. The number of words that can be formed using all the letters of the word "KANPUR" when the vowels are in even places is

1) 144 2) 36 3) 24 4) 48

21. The letters of the word "LOGARITHM" are arranged in all possible ways. The number of arrangements in which the relative positions of the vowels and consonants are not changed is

1) 4320 2) 720 3) 4200 4) 3420

22. The number of ways one can arrange words with the letters of the word "MADHURI" so that always vowels occupy the beginning, middle and end places is

1) 7! 2) ${}^3C_3 \cdot {}^4P_3$ 3) 3 . ${}^4C_4$ 4) 3! 4!

23. The letters of the word "FLOWER" are taken 4 at a time and arranged in all possible ways. The number of arrangements which begin with 'F' and end with 'R' is

1) 20 2) 18 3) 14 4) 12

24. The number of permutations that can be made from the letters of the word "SUNDAY" without beginning with 'S' or without ending with 'Y' is

1) 696 2) 624 3) 604 4) 504

25. Ten guests are to be seated in a row of which three are ladies. The ladies insist on sitting together while two of the gentlemen refuse to take consecutive seats. In how many ways can the guests be seated?

1) 2399976 2) 21844 3) 630624 4) 181440

26. The number of ways in which 6 Boys and 5 Girls can sit in a row so that all the girls maybe together is

1) 6! 5! 2) 6! ${}^7P_5$ 3) $(6!)^2$ 4) 7! 5!

27. The number of ways in which 5 Boys and 5 Girls can be arranged in a row so that no two girls are together is

1) 10! 2) 5! 6! 3) $(5!)^2$ 4) $2(5!)^2$

28. The number of ways can 4 men, 3 boys, 2 women be seated in a row so that the men, the boys and the women are not seperated is

1) 4! 3! 2! 2) $(4!)^2$ 3! 2!
3) 4! $(3!)^2$ 2! 4) 4! 3! $(2!)^2$

29. A, B, C are three persons among 7 persons who speak at a function. The number of ways in which it can be done if 'A' speaks before 'B' and 'B' speaks before 'C' is

1) 820 2) 830 3) 840 4) 850

30. The number of ways in which 20 different white balls and 19 different black balls be arranged in a row, so that no two balls of the same colour come together is

1) 20! ${}^{21}P_{19}$ 2) 20!19! 3) $(20!)^2$ 4) $(21)!\,{}^{20}C_{19}$

31. There are 10 white and 10 black balls marked 1,2,3 ..... 10. The number of ways in which we can arrange these balls in a row in such a way that neighbouring balls are of different colours is

1) 10! 9! 2) 20! 3) $(10!)^2$ 4) $2(10!)^2$

32. The number of ways in which 10 books can be arranged in a row such that two specified books are side by side is

1) $\frac{10!}{2!}$ 2) 9! 3) 9! 2! 4) $\frac{9!}{2!}$

33. The number of ways in which the time table for Monday be completed if there must be 5 lessons that day (Algebra, Geometry, Calculus, Trigonometry, Vectors) and Algebra and Geometry must not immediately follow each other are

1) 72 2) 5! 3) $3 \times 5!/2$ 4) 6!

34. If 'a' denotes the number of permutations of $x+2$ things taken all at a time, b the number of permutation of x things taken 11 at a time and c the number of permutations of $x-11$ things taken all at a time such that $a = 182bc$, then the value of x is

1) 15 2) 12 3) 10 4) 18

35. The number of four digit numbers that can be formed with 0, 1, 2, 3, 4, 5 is

1) ${}^6P_4$ 2) 5 . ${}^6P_3$
3) ${}^6P_4$ - ${}^5P_4$ 4) ${}^6P_4$ - ${}^5P_3$

36. The number of four digit odd numbers that can be formed with 1, 2, 3, 4, 5, 6, 7, 8, 9 is

1) 4 . ${}^8P_3$ 2) 5 . ${}^8P_3$ 3) 4 . ${}^7P_3$ 4) 5 . ${}^7P_3$

37. The number of four digit odd numbers that can be formed so that no digit being repeated in any number is

1) 2240 2) 2420 3) 2440 4) 2520

38. The number of four digit even numbers that can be formed with 0, 1, 2, 3, 7, 8 is
1) 180 2) 175 3) 160 4) 156

39. The number of numbers of 9 different non-zero digits such that all the digits in the first four places are less than the digit in the middle and all the digits in the last four places are greater than the digit in the middle is
1) 2(4!) 2) $(4!)^2$ 3) 8! 4) $2\times(\lfloor 4)^2$

40. Number of 6-digit telephone numbers, which can be constructed with digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, if each number starts with 35 and no digit appears more than once is
1) 1680 2) 8! 3) 6! 4) 6.6!

PROBLEMS ON DIVISIBILITY

41. If repetitions are not allowed, the number of numbers consisting of 4 digits and divisible by 5 and formed out of 0, 1, 2, 3, 4, 5, 6 is
1) 220 2) 240 3) 370 4) 588

42. The number of seven digit numbers divisible by 9 formed with digits 1, 2, 3, 4, 5, 6, 7, 8, 9 without repetition is
1) 7! 2) ${}^9P_7$ 3) 3(7!) 4) 4(7!)

43. A five-digit number divisible by 6 is to be formed by using 0, 1, 2, 3, 4, 5 without repetition. The number of ways in which this can be done is
1) 60 2) 48 3) 108 4) 216

44. A 5 digit number divisible by 3 is to be formed using the digits 0, 1, 2, 3, 4, 5 without repetition. The total number of ways this can be done is
1) 120 2) 96 3) 216 4) 220

45. The number of 5 digited numbers which are not divisible by 5 and which contains of 5 odd digits is
1)96 2) 120 3) 24 4) 32

46. The sum of integers from 1 to 100 that are divisible by 2 or 5 is
1) 3000 2) 3050 3) 3600 4) 3250

47. The number of five digit numbers that can be formed with 0, 1, 2, 3, 5 which are divisible by 25 is
1) 42 2) 24 3) 10 4) 38

48. The number of ways in which we can arrange the digits 1, 2, 3....9 such that the product of five digits at any of the five consecutive positions is divisible by 7 is
1) 7! 2) ${}^9P_7$ 3) 8! 4) $5(7!)$

SUM OF THE NUMBERS

49. The sum of the digits at the ten's place of all the numbers formed with the digits of 3, 4, 5, 6 taken all at a time is
1) 432 2) 108 3) 36 4) 18

50. The sum of the value of the digits at the ten's place of all the numbers formed with the help of 3, 4, 5, 6 taken all at a time is
1) 1080 2) 4320 3) 360 4) 180

51. The sum of all the four digit numbers that can be formed with 0, 2, 3, 5 is
1) 66660 2) 66480 3) 64440 4) 65520

PERMUTATIONS WHEN REPETITIONS ARE ALLOWED & LIKE THINGS

52. The number of permutations that can be made out of the letters of the word "ENTRANCE" so that the two 'N's are always together is
1) $\frac{7!}{(2!)^2}$ 2) 7! 3) $\frac{7!}{2!}$ 4) $\frac{7!}{(2!)^3}$

53. The number of permutations that can be made by using all the letters of the word TATATEACUP that start with A and end with U is
1) 5040 2) 8! 5! 3! 3! 3) 3360 4) 360

54. All the letters of the word EAMCET are arranged in all possible ways. The number of such arrangements in which no two vowels are adjacent to each other is
1) 36 2) 54 3) 72 4) 144

55. If the number of ways in which n different things can be distributed among n persons so that at least one person does not get any thing is 232. Then n is equal to
1) 3 2) 4 3) 5 4) 6

56. Number of ways of permuting the letters of the word "ENGINEERING" so that the order of the vowels is not changed is
1) ${}^{11}P_5$ 2) $\frac{11!}{5!}$ 3) $\frac{11!}{2!5!}$ 4) $\frac{{}^{11}P_5}{2}$

57. The number of different numbers that can be formed by using all the digits 1, 2, 3, 4,3, 2, 1 so that odd digits always occupy the odd places is
1) 6 2) 72 3) 60 4) 18

58. A three digit number n is such that the last two digits of it are equal and different from the first. The number of such n's is
1) 64 2) 72 3) 81 4) 900
59. The number of three digit numbers having only two consecutive digits identical is
1) 153 2) 162 3) 168 4) 163
60. The number of 'n' digit numbers such that no two consecutive digits are same is
1) 9! 2) n^9 3) 9^n 4) 9n
61. The number of five digit numbers formed using the digits 0, 2, 2, 4, 4, 5 which are greater than 40,000 is
1) 84 2) 90 3) 72 4) 60
62. The number of numbers greater than or equal to 1000 but less than 4000 that can be formed with 0, 1, 2, 3, 4 so that any digit may be repeated is
1) 374 2) 375 3) 120 4) 360
63. The sum of all 3 digited numbers that can be formed from the digits 1 to 9 and when the middle digit is a perfect square is (repetitions are allowed)
1) 1,34,055 2) 2,70,540
3) 1,70,055 4) 2,34,520

PROBLEMS ON NUMBER OF FUNCTIONS

64. Number of functions from Set-A containing 5 elements to a set-B containing 4 elements is
1) 5^4 2) 4^5 3) 4! 4) 5!
65. The number of one one functions that can be defined from $A=\{a,b,c\}$ into $B=\{1,2,3,4,5\}$ is
1) ${}^5P_3$ 2) ${}^5C_3$ 3) 5^3 4) 3^5
66. The number of many one functions from $A=\{1,2,3\}$ to $B=\{a,b,c,d\}$ is
1) 64 2) 24 3) 40 4) 0
67. The number of onto functions that canbe defined from $A=\{a,b,c,d,e\}$ to $\{1,2\}$ is
1) 30 2) 0 3) 60 4) 32
68. The number of one one onto functions that can be defined from $A=\{a,b,c,d\}$ into $B=\{1,2,3,4\}$ is
1) 12 2) 24 3) 18 4) 26
69. The number of constant mappings from $A=\{1,2,3,4............,n\}$ to $B=\{a,b\}$ is
1) n! 2) n 3) 2 4) 2^n

PROBLEMS BASED ON RANK MODEL

70. The rank of the word "MADHUR" when arranged in dictionary order is
1) 362 2) 360 3) 358 4) 356
71. The letters of the word "DANGER" are permuted in all possible ways and the words thus formed are arranged as in a dictionary. The rank of the word "DANGER" is
1) 132 2) 133 3) 134 4) 135
72. The letters of the word MASTER are permuted in all possible ways and the words thus formed are arranged as in a dictionary. The rank of the word STREAM is
1) 597 2) 480 3) 612 4) 285
73. If the letters of the word "NARAYANA" are arranged as in a dictionary then the Rank of the given word is
1) 511 2) 512 3) 513 4) 514
74. The letters of the word CRICKET are permuted in all possible ways and the words thus formed are rearranged as in a dictionary. The rank of the word CRICKET is
1) 243 2) 452 3) 531 4) 729
75. All the numbers that can be formed using the digits, 1, 2, 3, 4 are arranged in the increasing order in magnitude. The rank of the number 3241 is
1) 56 2) 16 3) 55 4) 15
76. All the numbers that can be formed using the digits 1, 2, 3, 4, 5 are arranged in the decreasing order of magnitude. The rank of 34215 is
1) 58 2) 62 3) 96 4) 128

PROBLEMS BASED ON CIRCULAR PERMUTATIONS

77. A round table conference is to be held between 20 deligates of 20 countries. The number of ways in which they can be seated if two particular deligates are always to sit together is
1) 19! 2! 2) 18! 2! 3) 18! 4) 19!

78. The number of ways in which 5 men, 5 women and 12 children can sit around a circular table so that the children are always together is

1) 4! 4! 12! 2) 11! 12! 3) 10! 12! 4) $(12!)^2$

79. 20 persons are invited for a party. The different number of ways in which they can be seated at a circular table with two particular persons seated on a either side of the host is

1) 19! 2! 2) 18! 2! 3) 20! 2! 4) 18! 3!

80. The number of ways in which 4 men and 4 women are to sit for a dinner at a round table so that no two men are to sit together is

1) 576 2) 144 3) 36 4) 120

81. The number of ways in which 7 men can sit at a round table so that all shall not have the same neighbours in any two arrangements is

1) 360 2) 720 3) 700 4) 300

82. 7 women and 7 men are to sit round a circular table such that there is a man on either side of every women. The number of seating arrangements is

1) $(7!)^2$ 2) $(6!)^2$ 3) 6! 7! 4) 7!

83. The number of circular permutations of 7 dissimilar things taken 5 at a time is

1) 2520 2) 1260 3) 500 4) 504

84. The number of ways in which 8 red roses and 5 white roses of different sizes can be made out to form a garland so that no two white roses come together is

1) $\frac{8!}{2}\cdot {}^8P_5$ 2) $\frac{7!}{2}\cdot {}^8P_5$ 3) $\frac{7!}{2}\cdot {}^9P_5$ 4) $7!\,{}^4P_3$

85. The number of ways that a garland can be made out of 6 red and 4 white roses of different sizes, so that all the white roses come together is

1) 8640 2) 4320 3) 720 4) 360

PALINDROMES

86. Number of 5 digit palindromes is

1) 8100 2) 900 3) 90000 4) 9000

87. Number of 5 digit Even palindromes is

1)400 2) 500 3) 4000 4) 5000

LEVEL-I (C.W.) - KEY

1) 4	2) 4	3) 1	4) 2	5) 1	6) 1	7) 4
8) 1	9) 2	10) 2	11) 4	12) 4	13) 1	14) 1
15) 1	16) 3	17) 3	18) 3	19) 2	20) 1	21) 1
22) 4	23) 4	24) 1	25) 4	26) 4	27) 2	28) 3
29) 3	30) 2	31) 4	32) 3	33) 1	34) 2	35) 4
36) 2	37) 1	38) 4	39) 2	40) 1	41) 1	42) 4
43) 3	44) 3	45) 1	46) 2	47) 3	48) 3	49) 2
50) 1	51) 3	52) 3	53) 3	54) 3	55) 2	56) 4
57) 4	58) 3	59) 2	60) 3	61) 2	62) 2	63) 1
64) 2	65) 1	66) 3	67) 1	68) 2	69) 3	70) 1
71) 4	72) 1	73) 1	74) 3	75) 1	76) 1	77) 2
78) 3	79) 2	80) 2	81) 1	82) 3	83) 4	84) 2
85) 1	86) 2	87) 1				

LEVEL-I (C.W.) - HINTS

1. By fundamental theorem of Multiplication= 5x8
2. By fundamental theorem of Multiplication= 4x3x2
3. By fundamental theorem of addition
   10 + 8 = 18
4. ${}^nP_r + r.{}^nP_{r-1} = {}^{(n+1)}P_r$
5. If $\frac{{}^nP_{r-1}}{a} = \frac{{}^nP_r}{b} = \frac{{}^nP_{r+1}}{c} \Rightarrow b^2 = a(b+c)$
   $\Rightarrow 9 = a\times 7$
6. $\frac{\frac{(2n+1)!}{(n+2)!}}{\frac{(2n-1)!}{(n-1)!}} = \frac{3}{5} \Rightarrow n = 4$
7. $\frac{n!}{(n-7)!} = 42.\frac{n!}{(n-5)!} \Rightarrow n = 12$
8. 32760 = 15 x 14 x 13 x 12 $\Rightarrow r = 4$
9. ${}^nP_r = 5040 = 10\times 9\times 8\times 7 \Rightarrow r = 4; n = 10$
10. $\sum_{r=1}^{10} r.r! = \sum_{r=1}^{10}(r+1-1).r! = \sum_{r=1}^{10}\left[(r+1)!-r!\right] = 11!-1$
11. n . n! = [(n+1) - 1] n! = (n+1)! - n!
    $\therefore$ 1+2! - 1! + ... + (n+1)! - n! = (n+1)!
12. k! 1! 2! 3! 4! 5! 6! 7! 8! 9! 10! 11! ...
    last 2 digits 01 02 06 24 20 20 40 20 80 00 00..
    $\therefore$ last two digits in above expression are same as the last two digits of $\sum_{k=1}^{9} k! = 13$.

13. $\dfrac{1!+2!+3!+4!}{20}$ gives a remainder 13

14. n! = 1.2.3.....n i.e. n! is divisible by any number between 2 to n. $\therefore$ n! + 1 when divided by any number between 2 to n leaves '1' as remainder

15. $\sum_{K=1}^{m}\left(K^2+1\right)K! = \sum_{K=1}^{m}\left(\left(K+1\right)^2-2K\right)K!$

$= \sum_{K=1}^{m}(K+1).(K+1)!-2\sum_{K=1}^{m}K.K!$

$= \left((m+2)!-2\right)-2\left((m+1)!-1\right)$

= m.(m+1)! = 1999.2000! $\Rightarrow$ m = 1999

16. ${}^4P_1+{}^4P_2+{}^4P_3+{}^4P_4=64$

17. Wearing of coats = ${}^4P_3$
Wearing of waist coats = ${}^5P_3$
Wearing of caps = ${}^6P_3$

18. $4!\times 6!$

19. ${}^6P_1+{}^6P_2+{}^6P_3+{}^6P_4+{}^6P_5+{}^6P_6$

20. 2 vowels occupy 3 even places to ${}^3P_2$ ways remaining 4 places ocupy by consonants in 4!
$\therefore$ Ans ${}^3P_2$ x 4!

21. LGRTHM; OAI - 6!.3!=4320

22. 3 vowels, 3 places = 3! remaining in 4!
Total = 3! 4!

23. ${}^4P_2$

24. $n\left(\overline{A}\cup\overline{B}\right)=n\left(\overline{A\cap B}\right) \quad =n(s)-n(A\cap B)$

25. $3!8!-3!2!7!$

26. $7!.5!$ (Taking all 5 girls together as one person)

27. 5 boys arranged in 5! Out of the 6 gaps 5 girls sit in ${}^6P_5$ way. $\therefore$ 5! ${}^6P_5$

28. 4m + 3b + 2w = 3 objects 3! 4! 3! 2!

29. ${}^7P_{7-3}={}^7P_4=840$

30. $20!.19!$ (Black balls in between white balls)

31. 10 white balls are arranged in 10! ways
10 black balls are arranged in 10! ways
Starting with white ball + starting with black balls = 10! x 10! + 10! x 10!

32. Two specified books = 1 object; they internally arranged in 2! So remaining 8 books + 1 object = 9 objects $\therefore$ 9! 2!

33. x V x C x T x
${}^4P_2$. 3!

34. $a=(x+2)!$; $b={}^xP_{11}$; $c=(x-11)!$

35. ${}^6P_4-{}^5P_3$ (deleting numbers starting with 0)

36. ${}^8P_3\times 5$

37. Last place in 5 ways (1,2,5,7, 9), first place in 8 ways, $= 8^2\times 7\times 5=2240$

38. If 0 is in units place no. of ways = ${}^5P_3=60$
If 2 or 8 is in units place no. of ways = $2\left({}^5P_3-{}^4P_2\right)=96$ Total : 60 + 96 = 156

39. Fix 5 at middle place, 1, 2, 3, 4 on left side and 6, 7, 8, 9 on right side can be arranged in $(4!)^2$ ways.

40. Fix. 35 in the begining two places. The remaining 4 places can be filled in ${}^8P_4$ ways

41. ${}^6P_3+{}^6P_3-{}^5P_2=220$

42. The number is divisible by 9 if sum of digits is divisible by 9. So the seven digit number should not contain (1, 8) or (2, 7) or (3, 6) or (4, 5) and the remaining digits can be arranged in 7! ways
$\therefore$ No. of such numbers is = 4 (7!)

43. Number divisible by 2 and 3 is divisible by 6
Case (i) delete 0
i) fix 2 in units place ---> 4!
ii) fix 4 in units place ---> 4!
Case (ii) detete 3
i) fix 0 in units place ---> 4!
ii) fix 2 in units place ---> 4! - 3!
iii) fix 4 in units place ---> 4! - 3!

44. If a number divisible by 3, its sum of digits divisibile by 3 {0, 1, 2, 4, 5} or {1, 2, 3, 4, 5}

45. $5!-4!=96$

46. Required sum = (2 + 4 + 6 + ... + 100)
+ (5 + 10 + 15 + ... + 100)
- (10 + 20 + ... + 100) = 3050

47. Last two digits are 50 or 25 i.e. 3! + (3!-2!) = 10

48. Product of any five consecutive digits is divisible by 7, only when '7' occupies the middle digit
$\therefore$ Remaining 8 places can be occupied in 8! ways

49. $3!(3+4+5+6)=108$

50. $3!(3+4+5+6).10=1080$

51. 3! x 1111 x 10 – 2! x 111 x 10 = 64440

52. TRAC; E E N N; $\dfrac{7!}{2!}$

53. $\frac{8!}{2!3!}$

54. EEA - VOWELS; MCT - CONSONENTS

$3!.\frac{{}^4P_3}{2!} = 72$

55. $n^n - n! = 232 \Rightarrow n = 4$

56. NNN GGR IIEEE $= \frac{{}^{11}P_6}{3!.2!} = \frac{{}^{11}P_5}{\underline{|2}}$, (3! & 2! in Dr for 3N's and 2G's)

57. Odd digits in odd places can be filled in $\frac{4!}{2!2!}$

and Even digits in even places can be filled in $\frac{3!}{2!}$

Therefore $\frac{4!}{2!2!}\frac{3!}{2!}$

58. 9 x 9 = 81 (first places in 9 ways and II and III places together in 9 ways)

59. Two non zero digits can be selected in ${}^9C_2 = 36$

$\Rightarrow 36 \times 4 = 144$ Eg : 1, 2 {112, 122, 211, 221}
One zero and one non zero in 9 ways
= 9 x 2 = 18 Eg : 0, 1 {110, 100}
Total = 144 + 18 = 162

60. There are {0, 1, 2,.....10}, 10 different digits of which I place is filled in 9 ways,
II place in 9 ways
(since 0 can be included and first digit is not

61. 4 x x x x x x $= \frac{\underline{|5}}{\underline{|2}}$ =60; 5 x x x x x x $= \frac{\underline{|5}}{(\underline{|2})^2} = 30$

62. First place in 3 ways by 1 or 2 or 3 $\Rightarrow$ 3 x 5^3 = 375

63. The value contributed by given numbers in units place = 9 x 3 x (1+2+.........+9)
Similarly in ten's place = 9 x 9 x (10+40+90)
Similarly in hundreds's place = 3 x 9 x (100+200+......+ 900)
$\therefore$ Total $= 1,34,055$

64. n^m

65. ${}^nP_m$

66. Total Functions - Number of one-one functions

67. $2^m - 2$

68. n!

69. n(B)

70. MADHUR : ADHMRU
3.5! + 0.4! + 0.3! + 0.2! + 1.1! + 1 = 362

71. A D E G N R ; DANGER
$1.5! + 0.4! + 2.3! + 1.2! + 0.1! + 1 = 135$

72. $4(5!) + 4(4!) + 3(3!) + 2! + 1 = 597$

73. A A A A N N R Y
N A R A Y A N A
$\frac{7! \times 4}{4! \times 2!} + \frac{4 \times 5!}{3!} + \frac{3 \times 3!}{2!} + 1 \times 1 + 1 = 511$

74. C C E I K R T
CRICKET
4(5!) + 2(4!) + 2! + 1 = 531

75. No. of single digit numbers = 4
+ No. of 2 digit numbers = ${}^4P_2$
+ No. of 3 digits numbers = ${}^4P_3$
+ above problem

76.
3	4	2	1	5
5	4	3	2	1

2(4!) + 1(3!) + 1(2!) + 1(1!) + 1 = 58

77. 18!. 2!

78. 10!.12!

79. 2.18!

80. 3!. 4!

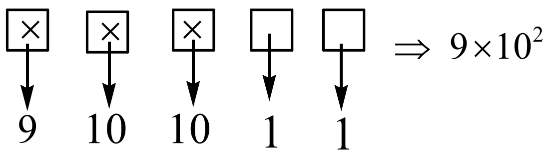
81. $\frac{6!}{2!} = 360$

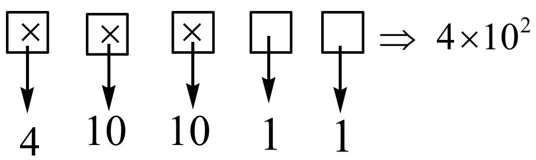
82. 6!. 7!

83. ${}^7C_5.4! = \frac{7!}{5!.2!} \times 4! = 504$

84. $\frac{7!}{2!}.{}^8p_5$

85. $\frac{6!.4!}{2!} = \frac{17280}{2!} = 8640$

86. 

87. 

LEVEL -I (H.W)

FUNDAMENTAL PRINCIPLE

1. There are 5 doors to a lecture hall. The number of ways that a student can enter the hall and leave it by a different door is
1) 20 2) 16 3) 19 4) 21

2. 15 buses fly between Hyderabad and Tirupati. The number of ways can a man go to Tirupati from Hyderabad by a bus and return by a different bus is
1) 15 2) 150 3) 210 4) 225

PROBLEMS BASED ON ${}^{n}P_{r}$ FORMULA

3. If ${}^{(n+1)}P_5 : {}^{n}P_5 = 3:2$ then n is
1) 8 2) 14 3) 10 4) 6

4. ${}^{15}P_8 = A + 8.{}^{14}P_7 \Rightarrow A =$ [EAM 2011]
1) ${}^{14}P_6$ 2) ${}^{14}P_8$ 3) ${}^{15}P_7$ 4) ${}^{15}P_9$

5. If ${}^{(m+n)}P_2 = 90,\ {}^{(m-n)}P_2 = 30$ then (m, n) =
1) (10, 3) 2) (9, 2) 3) (8, 2) 4) (7, 3)

6. There are 50 stations on a railway line. The number of different kinds of single 2nd class tickets must be printed so as to enable a passenger to travel from one station to another station is
1) ${}^{50}C_2$ 2) 2500 3) 2450 4) 50!

7. If ${}^{n}P_r = 3024$ then (n, r) =
1) (8, 4) 2) (8, 3) 3) (9, 3) 4) (9, 4)

PROBLEMS BASED ON REMAINDER & OTHER MODELS

8. The product 5. 6. 7. ....(r + 4) is divisible by
1) (r - 2)! 2) (r - 1)! 3) r! 4) (r + 1)!

9. $\frac{1}{3.1!} + \frac{1}{4.2!} + \frac{1}{5.3!} + ...\infty =$
1) 1 2) 2 3) 1/2 4) 3

PERMUTATIONS OF DISSIMILAR THINGS

10. In a class of 10 students there are 3 girls. The number of ways they can be arranged in a row, so that no two girls are consecutive is k.8!, where k =
1) 42 2) 12 3) 24 4) 36

11. The number of ways of arranging 6 players to throw the cricket ball so that the oldest player may not throw first is
1) 120 2) 600 3) 720 4) 175

12. The number of times the digit '5' will be written while listing the integers from 1 to 1000 is
1) 271 2) 272 3) 300 4) 285

13. 10 men and 6 women are to be seated in a row so that no two women sit together, the number of ways they can be seated is [EAM 2013]
1) 11!.10! 2) $\frac{11!}{6!5!}$ 3) $\frac{10!9!}{5!}$ 4) $\frac{11!10!}{5!}$

14. 9 balls are to be placed in 9 boxes, and 5 of the balls can not fit into 3 small boxes. The number of ways of arranging one ball in each of the boxes is [EAM 2008]
1) 18720 2) 18270 3) 17280 4) 12780

15. The total number of signals that can be made with 7 different coloured flags is
1) 128 2) 7 3) 127 4) 13699

16. In a class of 10 students, there are 3 girls A, B, C. The number of different ways that they can be arranged in a row such that no two of the three girls are consecutive is
1) 10! 2) 7! 8!
3) 7! 8!/5! 4) 7!8!3!/51

17. The number of ways that 5 students can be sit in a row so that the tallest and shortest may not come together is
1) 48 2) 24 3) 72 4) 120

18. $S_1, S_2, ..., S_{10}$ are the speakers in a conference. If S_1 addresses only after S_2, then the number of ways the speakers address is
1) 10! 2) 9!
3) 10 x 8! 4) (10!)/2

19. The different words that can be formed by using the letters of the word BHARAT so that these words will not contains B and H together is
1) 360 2) 120 3) 240 4) 480

20. The number of arrangements that can be made by using all the letters of the word 'MATRIX' so that the vowels may be in the even places is
1) 144 2) 2880 3) 720 4) 5760

21. The total number of 9 digit numbers which have all different digits is
1) 9.9! 2) 10! 3) ${}^{10}P_9$ 4) 9^9

22. Number of 2 digit even numbers that can be formed from the digits 1, 2, 3, 4 and 5 when repetitions are not allowed is
1) 8 2) 5! 3) 26 4) 16

23. The number of numbers greater than 23000 can be formed from 1, 2, 3, 4, 5 is
1) 120 2) 96 3) 91 4) 90

24. Two persons entered a Railway compartment in which 7 seats were vacant. The number of ways in which they can be seated is
1) 30 2) 42 3) 720 4) 360

25. There are 8 floors on a building including the ground floor. If 4 persons enter the lift in the ground floor the number of ways in which they get down the lift if no two persons come out of the lift at the same floor is
1) ${}^{7}C_{4}$ 2) 7^{4} 3) ${}^{7}P_{4}$ 4) 4^{7}

26. The number of four digit even numbers that can be formed with 1, 2, 3, 4, 5, 6 is
1) 125 2) 180 3) 120 4) 360

27. The number of ways in which the letters of the word 'INSURANCE' be arranged so that the vowels are never be separated is
1) 4320 2) 8640 3) 21600 4) 10300

28. The prismatic colours are arranged in a row. The number of arrangements in which Red and Blue come together is
1) 720 2) 360 3) 2880 4) 1440

PROBLEMS ON DIVISIBILITY

29. The number of 6 digited numbers which are not divisible by 5 that can be formed with the digits 4, 5, 6, 7, 8, 9 is
1) 120 2) 720 3) 600 4) 840

30. The number of four digited numbers which are not divisible by 2 that can be formed from the digits 2, 3, 4, 5, 6 is
1) 37 2) 48 3) 45 4) 34

31. The number of 4 digit numbers formed using the digits 1, 2, 3, 4, 5, 6, 7 which are divisible by 4 is
1) 100 2) 150 3) 200 4) 250

SUM OF THE NUMBERS

32. The sum of all possible numbers greater than 2000 formed by using the digits 2, 3, 4, 5 is
1) 93324 2) 96324 3) 92324 4) 3668

PERMUTATIONS WHEN REPETITIONS ARE ALLOWED & LIKE THINGS

33. In a town the car plate numbers contain only three or four digits, not containing the digit 0. The maximum number of cars that can be numbered is
1) 6480 2) 7290 3) 5862 4) 3528

34. The number of 6 digit numbers in which all the odd digits and only odd digits appear, is
1) $\frac{5}{2}(6!)$ 2) $6!$ 3) $\frac{1}{2}(6!)$ 4) $\frac{5!}{2}$

35. Four dice are rolled. The number of possible outcomes in which atleast one die shows 2 is
1) 1296 2) 625 3) 615 4) 671

36. The number of four digit numbers that can be formed with the digits 1, 2, 3, 4 and 5 in which atleast two digits are identical is
1) $4^{5}-5!$ 2) $5^{4}-5!$ 3) $4^{5}-5$ 4) $5^{4}-5$

37. Number of arrangements of the letters of the word 'BANANA' in which two N's do not appear adjacently is
1) 40 2) 60 3) 80 4) 100

38. The number of four digit telephone numbers having at least one of their digits repeated is
1) 4960 2) 4959 3) 10000 4) 5040

39. A guardian with 5 wards wishes everyone of them to study either graduation in Arts or Science or Engineering. The number of ways he can make up his mind with regard to the education of his wards if everyone of them is fit for any of those branches of study is
1) 5! 2) 243 3) 125 4) 325

40. The number of ways in which 7 distinct toys can be distributed among 3 children when each child is eligible to take all the toys is
1) 3^{7} 2) 7^{3} 3) ${}^{7}P_{3}$ 4) ${}^{7}C_{3}$

41. The number of odd numbers lying between 40000 and 70000 that can be made from the digits 0, 1, 2, 4, 5, 7 if digits can be repeated any number of times is
1) 1125 2) 1296 3) 766 4) 655

42. A letter lock consists of 4 rings each marked with 10 different letters, the number of ways in which it is possible to make an unsuccessful attempts to open the lock is
1) 8999 2) 9999 3) 4799 4) 5689

43. 5 dice are thrown then the number of out comes in which there is atleast one three is
1) 6^5-1 2) 6^5-5^5 3) 6^5-6.5^4 4) 5^5-1

44. A library has 'a' copies of one book, 'b' copies of two books, 'c' copies of each of three books and single copy of d books. The total number of ways in which these books can be distributed is

1) $\frac{(a+2b+3c+d)!}{a!b!c!}$ 2) $\frac{(a+b+c+d)!}{a!b!c!}$

3) $\frac{(a+2b+3c+d)!}{a!(b!)^2(c!)^3}$ 4) $\frac{(a+2b+3c+d)!}{a!(b!)^3(c!)^2}$

45. The number of ways that the letters of the word "NELLORE" be arranged so that 'N' and 'R' are always together is
1) 360 2) 720 3) 1260 4) 2520

46. The number of ways in which all the letters of the word "INTEGRATION" can be arranged so that all vowels are always in the begining of the word is

1) $\frac{6!\,5!}{(2!)^3}$ 2) $\frac{7!\,4!}{2!}$ 3) 7!. 4! 4) $\frac{7!}{(2!)^2}$

47. Number of arrangements that can be formed using the letters A thrice, B twice and C once is
1) 60 2) 120 3) 90 4) 59

PROBLEMS ON NUMBER OF FUNCTIONS

48. The number of ways of 3 scholarships of unequal value be awarded to 17 candidates, such that no candidate gets more than one scholarship is

1) ${}^{17}C_3$ 2) 17^3 3) 3^{17} 4) ${}^{17}P_3$

49. Each of the 5 questions in multiple choice test has 4 possible answers. The number of different sets of possible answers is
1) 1023 2) 5^4-1 3) 1024 4) 256

PROBLEMS BASED ON RANK MODEL

50. The letters of the word 'ZENITH' are permuted in all possible ways and the words thus formed are arranged as in a dictionary. The rank of the word 'ZENITH' is
1) 616 2) 618 3) 597 4) 5930

51. The letters of the word 'MIRROR' are taken and permuted in all possible ways. The words thus formed are arranged as in a dictionary then the rank of the given word is
1) 22 2) 23 3) 24 4) 25

52. The letters of the word 'AEIOU' are permuted in all possible ways and the words thus formed are arranged as in a dictionary, the rank of the word "UIOEA" is
1) 118 2) 116 3) 114 4) 120

53. Four digit numbers formed with 1, 2, 4, 6, 8 without repetition are formed in ascending order, then the rank of 4618 is
1) 31 2) 62 3) 124 4) 248

PROBLEMS BASED ON CIRCULAR PERMUTATIONS

54. The number of ways of arranging 8 men and 4 women around a circular table such that no two women sit together is

1) 8! 2) 4! 3) 8! 4! 4) 7! ${}^8P_4$

55. The number of ways that the chief minister and the 14 ministers of our state can sit at a round table for a conference so that chief minister can occupy the fixed seat is
1) 13! 2) 14! 3) 14!/2 4) 15!

PALINDROMES

56. The no. of 6 letter palindromes that can be formed using the letters of the word 'EQUATION' are

1) 8^6 2) 8^3 3) 8^5 4) 8^4

57. The number of palindromes with 5 digits that can be formed using the digits 0, 1, 2, 3, 4, 5 are

1) 5×6^2 2) 5×6^3 3) 5×6^4 4) 5^3

LEVEL-I (H.W.) - KEY

1) 1	2) 3	3) 2	4) 2	5) 3	6) 3	7) 4
8) 3	9) 3	10) 1	11) 2	12) 3	13) 4	14) 3
15) 4	16) 3	17) 3	18) 4	19) 3	20) 1	21) 1
22) 1	23) 4	24) 2	25) 3	26) 2	27) 2	28) 4
29) 3	30) 2	31) 3	32) 1	33) 2	34) 1	35) 4
36) 2	37) 1	38) 2	39) 2	40) 1	41) 2	42) 2
43) 2	44) 3	45) 1	46) 1	47) 1	48) 4	49) 3
50) 1	51) 2	52) 3	53) 2	54) 4	55) 2	56) 2
57) 1						

LEVEL-I (H.W.) - HINTS

1. $5 \times 4 = 20$
2. $15 \times 14 = 210$
3. Expand
4. ${}^{n}P_{r} = {}^{(n-1)}P_{r} + r.{}^{(n-1)}P_{r-1}$; $n = 15, r = 8$

   $\therefore A = {}^{(n-1)}P_{r} = {}^{14}P_{8}$
5. ${}^{(m+n)}P_{2} = {}^{10}P_{2}$; ${}^{m-n}P_{2} = {}^{6}P_{2}$

   $m + n = 10;\ m - n = 6$
6. 50×49
7. $3024 = 72 \times 42 = 9 \times 8 \times 7 \times 6 = {}^{9}P_{4}$
8. Product of r consecutive natural numbers is divisible by r!.
9. $\frac{1}{3.1!} + \frac{1}{4.2!} + \frac{1}{5.3!} + ...\infty$

   $= \frac{3-1}{3!} + \frac{4-1}{4!} + \frac{5-1}{5!} + ...$

   $= \frac{1}{2!} - \frac{1}{3!} + \frac{1}{3!} - .... = \frac{1}{2}$
10. $7! \times {}^{8}P_{3} = k \times 8!$
11. $5 \times 5! = 600$
12. $\_\,\_\,5 = 10^2$; $\_\,5\,\_ = 10^2$; $5\,\_\,\_ = 10^2$

    $\therefore$ required times $= 300$.
13. $10!\ {}^{11}P_{6} = \frac{10!11!}{5!}$
14. ${}^{6}P_{5} \times 4! = 720 \times 24 = 17280$
15. ${}^{7}P_{1} + {}^{7}P_{2} + {}^{7}P_{3} + {}^{7}P_{4} + {}^{7}P_{5} + {}^{7}P_{6} + {}^{7}P_{7}$
16. Given students other than A, B, C can be arranged in a row in 7! ways. There are 8 places to sit the three girls A, B, C. It can be done in ${}^{8}P_{3}$ ways.

    The arrangements are $7! \times {}^{8}P_{3}$
17. $5! - 4! \times 2!$
18. $\frac{10!}{2!}$
19. $\frac{6!}{2!} - \frac{5!}{2!} \times 2!$
20. ${}^{3}P_{2} \times 4!$
21. $9 \times 9!$
22. ${}^{4}P_{1} \times 2 = 8$
23. $1 \times 3 \times 3! + 3 \times 4!$
24. ${}^{7}P_{2} = 7 \times 6 = 42$
25. ${}^{8-1}P_{4} = {}^{7}P_{4}$
26. ${}^{5}P_{3} \times 3 = 180$
27. $\frac{(IUAE)}{1\ oject} + \frac{N,S,R,N,C}{5\ objects} = 6\ objects$

    $\therefore \frac{6!}{2!} \times 4!$
28. Number of colours = 7
    Consider Red and Blue as one unit.
    It is 2! ways with this and remaining are 6.
    They are in 6! ways $\therefore$ 6! 2! = 1440.
29. Required numbers $= 5 \times 5! = 5 \times 120 = 600$
30. Required numbers $= 2 \times {}^{4}P_{3} = 48$
31. If number divisible by 4, last two digits divisible by 4. Last two digits are 12, 16, 24, 32, 36, 52, 56, 72, 76

    No. of ways $= {}^{5}P_{2} \times 10 = 200$.
32. Sum $= (2+3+4+5)(3!)(1111) = 93324$.
33. $9^3 + 9^4 = 7290$
34. Clearly, one of the odd digits 1, 3, 5, 7, 9 will be repeated. The number of selections of the sixth digit $= {}^{5}C_{1} = 5$. Required, number of numbers $= 5 \times \frac{6!}{2!}$
35. $6^4 - 5^4 = 671$
36. $5^4 - {}^{5}P_{4}$
37. $\frac{4!}{3!} \times \frac{{}^{5}P_{2}}{2!}$
38. The number of four digit telephone numbers when each digit may be repeated any number of times $(10^4 - {}^{10}P_4) - 1$. ($\because$ 0000 is not a telephone number)
39. 3^5
40. 3^7
41. $2 \times 6 \times 6 \times 6 \times 3$.
42. $10^4 - 1$
43. $6^5 - 5^5$

44. $\dfrac{(1\times a+2\times b+3\times c+1\times d)!}{a!.(b!)^2.(c!).(1!)}$

45. $\dfrac{NR}{1}+\dfrac{EELLO}{t}$

$\therefore \dfrac{6!}{2!\,2!}\times 2!$

46. First arrange vowels begining, then arrange remainding letters.

47. $\dfrac{(3+2+1)!}{3!\,2!\,1!}$

48. Required number of ways are $17\times16\times15$

49. 4^5

50. Rank is

$5(5!)+0.4!+2(3!)+1\times2!+1\times1!+1=616$

51. M I R R O R

I M O R R R

$1.\dfrac{5!}{3!}+1.\dfrac{3!}{3!}+1.\dfrac{2!}{2!}+0.1!+1=23$

52. AEIOU : UIOEA

4.4! + 2.3! + 2.2! + 1.1! + 1

96 + 12 + 4 + 2 = 114.

53. 4 digit numbers begin with 1 = ${}^4P_3=24$

4 digit numbers begin with 2 = ${}^4P_3=24$

4 digit numbers begin with 41 = ${}^3P_2=6$

4 digit number begin with 42 = ${}^3P_2=6$

The next numbers are 4612, 4618

$\therefore$ rank = 24 + 24 + 6 + 6 + 1+1 = 62

54. $7!\ {}^8P_4$

55. $(15-1)!=14!$

56. n = 8, r = 6 (even)

$\therefore$ required palindromes = $n^{r/2}=8^3$

57. Given digits 0, 1, 2, 3, 4, 5

n = 6, r = 5 (odd)

$\underset{(5)}{\overline{}}\ \underset{(6)}{\overline{}}\ \underset{(6)}{\overline{}}\ \underset{(1)}{\overline{}}\ \underset{(1)}{\overline{}}$

$\therefore$ required palindromes $=(n-1)n^{\frac{r-1}{2}}$

($\because$ 0 including).

$=5\times6^2$

LEVEL-II (C.W)

PROBLEMS BASED ON ${}^nP_r$

1. Six examination papers are to be set in a certain order not to be disclosed. It is discovered that one order has been leaked out. The number of ways that their order can be changed is

1) ${}^6P_1$ 2) ${}^6C_1$ 3) 216 4) 719

2. 9 articles are to be placed in 9 boxes one in each box 4 of them are too big for three of the boxes. The number of possible arrangements is

1) 9! 2) 5! 4! 3) ${}^6P_4\times5!$ 4) 5! 6!

3. The number of different ways that three distinct rings can be worn to 4 fingers with atmost one ring in each of the fingers is

1) ${}^4P_3$ 2) ${}^5P_4$ 3) ${}^5P_3$ 4) ${}^4P_2$

'n^r' MODEL

4. There are 5 multiple choice questions in test. If the first three questions have 4 choices each and the next two have 5 choices each, the number of answers possible is

1) 1500 2) 1600 3) 1700 4) 1800

5. Sixteen men compete with one another in running, swimming and riding. How many prize lists could be made if there were altogether 6 prizes of different values, one for running, 2 for swimming and 3 for riding

1) 16 x 15 x 14 2) $16^3\times15^2\times14$

3) $16^3\times15\times14^2$ 4) $16^2\times15\times14$

6. The number of ways of wearing 6 different rings to 5 fingers is

1) 5^6 2) 6^5 3) 5^5 4) 6^6

PERMUTATIONS OF DISSIMILAR THINGS

7. The letters of the word "RANDOM" are arranged in all possible ways. The number of arrangements in which there are 2 letters between R and D is

1) 36 2) 48 3) 144 4) 72

8. A family of 4 brothers and 3 sisters is to be arranged in a row for a photograph. The number of ways in which they can be seated if all the sisters are to sit together is

1) 120 2) 240 3) 360 4) 720

9. If words are formed by taking only 4 at a time out of the letters of the word PHYSICAL, then the number of words in which Y occur is

1) ${}^{8}p_4$ 2) $\frac{7!}{3!}$ 3) $\frac{7!}{4!}$ 4) ${}^{7}c_4$

10. The number of permutations alltogether of n things when r specified things are to be in an assigned order, though not necessarily consecutive is

1) $\frac{n!}{(n-r)!}$ 2) $\frac{n!}{(n-r)!\,r!}$ 3) $\frac{n!}{r!}$ 4) (n-r)! . r!

11. The number of ways of permuting the letters of the word DEVIL so that neither D is the first letter nor L is the last letter is

1) 36 2) 114 3) 42 4) 78

12. The number of numbers lying between 10 and 1000 that can be formed with 0, 2, 3, 4, 5, 6 so that no digit being repeated in any number is

1) 100 2) 125 3) 120 4) 240

SUM OF THE NUMBERS

13. The sum of all the numbers that can be formed by taking all the digits from 2, 3, 4, 5 is

1) 93,324 2) 79,992 3) 66,66,600 4) 78,456

14. The sum of all 4 digited numbers that can be formed by taking the digits from 0, 1, 3, 5, 7, 9 is

1) 16,23,300 2) 16,32,300
3) 16,32,030 4) 16,33,200

15. The sum of all 4 digited even numbers that can be formed from the digits 1, 2, 3, 4, 5 is

1) 1,58,994 2) 1,59,984
3) 1,59,894 4) 1,59,884

PROBLEMS BASED ON REPETITIONS & SIMILAR THINGS

16. The number of numbers formed having not more than 6 digits using the numbers 3, 4, 5 when repetition is allowed is

1) 1092 2) 1090 3) 1085 4) 1064

17. The number of permutations of 25 dissimilar things taken more than 15 at a time when repetitions are allowed is (not exceeding 25)

1) $\frac{25}{24}\left(25^{25}-25^{15}\right)$ 2) $\frac{25}{24}\left(25^{25}-25^{10}\right)$

3) $\frac{25}{24}\left(25^{25}+25^{15}\right)$ 4) $\frac{25}{24}\left(25^{25}+25^{10}\right)$

18. A man has 3 servants. The number of ways in which he can send invitation cards to 6 of his friends through the servants is

1) 3^6 2) 6^3 3) $\frac{6!}{3!}$ 4) ${}^{6}P_3$

19. In the word 'ENGINEERING' if all 'E"s are not together and N's come together then number of permutations is

1) $\frac{9!}{2!\,2!}-\frac{7!}{2!\,2!}$ 2) $\frac{9!}{3!\,2!}-\frac{7!}{2!\,2!}$

3) $\frac{9!}{3!\,2!\,2!}-\frac{7!}{2!\,2!\,2!}$ 4) $\frac{9!}{3!\,2!\,2!}-\frac{7!}{2!\,2!}$

20. The number of permutations that can be made out of the letters of the word "MATHEMATICS" When all vowels come together is

1) $\frac{8!\,.\,4!}{2!}$ 2) $\frac{8!\,.\,4!}{(2!)^2}$ 3) $\frac{7!\,.\,4!}{2!}$ 4) 7! 4!

21. The number of permutation that can be made out of the letters of the word "MATHEMATICS". When no two vowels come together is

1) $7!\,{}^{8}P_4$ 2) $\frac{7!}{2!\,2!}\cdot{}^{8}p_4$ 3) $\frac{7!\,.\,{}^{8}p_4}{(2!)^3}$ 4) $7!\,\frac{{}^{8}p_4}{2!}$

22. The number of permutation that can be made out of the letters of the word "MATHEMATICS" When the relative positions of vowels and consonants remain unaltered is

1) 3 . 7! 2) 2 . 7! 3) 7! 4) 4 . 7!

23. The number of ways in which the letters of the word MULTIPLE be arranged without changing the order of the vowels is

1) 3360 2) 20160 3) 6720 4) 3359

24. If $n \in N$ and $300 < n < 3000$ and n is made of digits by taking from $0,1,2,3,4,5$ then greatest possible number of values of n is (repetition is allowed)

1) 260 2) 539 3) 320 4) 300

DEARRANGEMENT PROBLEMS

25. $f: A \to A,\ A=\{a_1, a_2, a_3, a_4, a_5\}$, then the number of one one functions so that $f(x_i) \neq x_i,\ x_i \in A$ is

1) 44 2) 88 3) 22 4) 20

26. The number of ways that all the letters of the word SWORD can be arranged such that no letter is in its original position is

1) 44 2) 32 3) 28 4) 20

27. A person writes letters to six friends and addresses corresponding envelopes let x be the number of ways so that atleast two of the letters are in wrong envelopes and y be the number of ways so that all the letters are in wrong envelopes then $x-y=$

1) 716 2) 454 3) 265 4) 0

CIRCULAR PERMUTATIONS

28. The no. of ways in which 6 gentlemen and 3 ladies be seated round a table so that every gentleman may have a lady by his side is ...

1) 1440 2) 720 3) 240 4) 480

29. The number of ways in which 7 men be seated at a round table so that two particular men are not side by side is

1) 2400 2) 120 3) 360 4) 480

30. Six persons A, B, C, D, E and F are to be seated at a circular table. The number of ways, this can be done if A must have either B or C on his right and B must have either C or D on his right is

1) 36 2) 12 3) 24 4) 18

OTHER MODELS

31. If $n=1!+4!+7!+.........+400!$ then ten's digit of 'n' is [EAM - 2010]

1) 5 2) 6 3) 7 4) 8

32. Let $a_n=\dfrac{10^n}{n!}$ for n = 1, 2, 3, ... Then the greatest value of n for which a_n is the greatest is [EAM 2010]

1) 11 2) 20 3) 10 4) 8

33. If $P(n,a)=P(n,b)$ where $a<b\le n$ (all are positive integers) and $a+b=Fn+G$ then $|F|+|G|=$

1) 2 2) 3 3) 4 4) 5

34. In a college of 300 students, every student read 5 newspapers and every newspaper is read by 60 students. The number of newspapers is

1) at least 30 2) at most 20
3) exactly 25 4) 24

35. The number of ways in which a TRUE or FALSE examination of 'n' statements can be answered on the asumption that no two consecutive questions are answered the same way is

1) 2^{n-1} 2) 2^n 3) 1 4) 2

36. In a steamer there are 12 stalls for animals and there are horses, cows and calves (not less than 12 each) ready to be shipped. In how many ways can the ship load be made such that any stall can't have morethan one animal

1) $3^{12}-1$ 2) 3^{12} 3) $(12)^3-1$ 4) $(12)^3$

LEVEL-II (C.W.) - KEY

1) 4	2) 3	3) 1	4) 2	5) 2	6) 1
7) 3	8) 4	9) 2	10) 3	11) 4	12) 2
13) 1	14) 4	15) 2	16) 1	17) 1	18) 1
19) 4	20) 4	21) 3	22) 1	23) 1	24) 2
25) 1	26) 1	27) 2	28) 1	29) 4	30) 4
31) 2	32) 3	33) 2	34) 3	35) 4	36) 2

LEVEL-II (C.W.) - HINTS

1. 6! - 1 = 719
2. 4 big articles placed in 6 big boxes in ${}^6P_4$ remaining 5 are placed in 5 boxes in 5! Ans : ${}^6P_4\times 5!$
3. $4\times3\times2$
4. First 3 Questions ---> 4.4.4
   Next 2 Questions ---> 5.5 Total = $4^3\cdot5^2$
5. No. of ways of giving '1' prize for running = 16
   No. of ways of giving '2' prizes for swimming = 16 x 15
   No. of ways of giving '3' prizes for riding = 16 x 15 x 14 = 16 x 16 x 15 x 16 x 15 x 14
   $=16^3\times15^2\times14$
6. Each ring can be worn to any one of 5 fingers in 5 ways
7. R, D can be arranged in 2! ways. Arrange two letters from 4 letters in ${}^4P_2$ ways and let total of this is one unit with this and remaining are 3! ways
   $\therefore$ Total = 2! . ${}^4P_2$. 3! ways
8. 3! 5!
9. PHYSICAL : Y can be arranged in 4 ways and remaining 3 letters from 7 letters are arranged in ${}^7P_3$ ways $\Rightarrow 4.{}^7P_3=\dfrac{7!}{3!}$

10. ${}^{n}C_r\,.\,(n-r)!$

11. Total arrangement with the letters of the word DEVIL = 5! = 120
No. of arrangement starting with D = 24 = 4!
No. of arrangement ending with L = 24 = 4!
No. of arrangement that begin with D and end with L is = 6
No. of arrangements required = 120-(24+24-6) = 78

12. 5 x 5 + 5 x 5 x 4

13. (2 + 3 + 4 + 5) (1111) 3! = 93,324

14. $(1+3+5+7+9)\left[{}^{5}P_3.(1111)-{}^{4}P_2\,(111)\right]$
= 16,33,200

15. ${}^{4}P_3.(2+4).1(\text{units place})+$
$\left(\frac{{}^{4}P_3}{4}(2+4)+\frac{{}^{4}P_3}{2}(1+3+5)\right)(1110) = 1{,}59{,}984$

16. $\frac{n\left(n^r-1\right)}{n-1}=\frac{3\left(3^6-1\right)}{3-1}=\frac{728\times 3}{2}=1092.$

17. From synopsis $\frac{n\left(n^n-n^r\right)}{n-1}$
here n = 25; r = 15 $\Rightarrow \frac{25\left(25^{25}-25^{15}\right)}{24}$

18. To send one invitation there are 3 chances.
So to send 6 invitations there are 3^6 chances.

19. NNN = One unit Remaining = 8 units. Total 9 units
$\therefore$ Total permutations - E's come together.

20. MMTTHCS; AAEI ; $\frac{8!\times 4!}{(2!)^3}=7!.4!$

21. $\frac{7!}{(2!)^2}.\frac{{}^{8}P_4}{2!}$

22. $\frac{7!.4!}{(2!)^3}=3.7!$

23. $\frac{{}^{8}P_5}{2!}$ x 1

24. $\left.\begin{matrix}3--\\4--\\5--\end{matrix}\right\}\Rightarrow 3\times 6^2$; $\left.\begin{matrix}1--\\2--\end{matrix}\right\}\Rightarrow 2\times 6^3$
No. of ways = $3\times 6^2+2\times 6^3-1=539$

25&26 $5!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right]=44$

27. $x=6!-1=719$
$y=6!\left[1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}\right]=265$
$x-y=719-265=454$

28. 5!. 3! . 2! = 1440

29. 6! - 5! x 2 = 480

30. When A has B or C to his right we have the order : AB or AC
When B has C or D to his right we have the order : BC or BD
Taking these two possibilities together, we must have ABC or ABD or AC and BD
For ABC, D, E, F to arrange around a circle no. of ways = 3! = 6
For ABD, C, E, F to arrange around a circle no. of ways = 3! = 6
For AC, BD, E, F to arrange around a circle no. of ways = 3! = 6
$\therefore$ Total no. of ways = 6 + 6 + 6 = 18

31. 1! + 4! + 7! + ... + 400! = 1 + 24 + 5040 + 3628800 + ... $\therefore$ ten's digit = 2 + 4 = 6

32. $a_{11}=\frac{10^{11}}{11!}=\frac{10}{11}a_{10}<a_{10}$
$a_{20}=\frac{10^{20}}{20!}=\frac{10^{10}}{11\times 12\times ....20}a_{10}<a_{10}$
$a_8=\frac{10^8}{8!}=\frac{9\times 10}{10^2}a_{10}<a_{10}$
$\therefore$ greatest value of n is 10

33. If $P(n,a)=P(n,b)$ where $a<b\le n$ (all are positive integers) and $a+b=Fn+G$
$P(n,a)=P(n,b)$
$\Rightarrow \frac{n!}{(n-a)!}=\frac{n!}{(n-b)!}\Rightarrow (n-a)!=(n-b)!$
and $n-a\ne n-b \;\Rightarrow n-a=1, n-b=0$
$\Rightarrow a+b=2n-1=Fn+G$
$\therefore |F|+|G|=|2|+|-1|=3$

34. The number of news papers = n, then 60n=300x5
$\therefore$ n = 25

35. True,False-1 way, False, True-1 Way
Number of ways=2

36. Each of 12 places can be filled by horse or cow or calf in three ways $\Rightarrow$ total ways = 3^{12}

LEVEL-II (H.W)

PROBLEMS BASED ON ${}^{n}P_{r}$

1. The number of different ways that three distinct rings can be worn to 4 fingers with atmost one ring in each of the fingers is
1) ${}^{4}P_{3}$ 2) ${}^{5}P_{4}$ 3) ${}^{5}P_{3}$ 4) ${}^{4}P_{2}$

2. On a new year day every student of a class sends a card to every other student. The postman delivers 600 cards. The number of students in the class are
1) 42 2) 34 3) 25 4) 52

'n^r' MODEL

3. The number of numbers greater than 1000 but not greater than 4000 that can be formed with the digits 0, 1, 2, 3, 4 repetition of digits being allowed is
1) 375 2) 376 3) 377 4) 378

4. An n-digit number is a positive number with exactly n digits. Nine hundred distinct n-digit numbers are to be formed using only three digits 2, 5 and 7. The smallest value of n for which this is possible is
1) 6 2) 7 3) 8 4) 9

5. The number of numbers less than 2000 that can be formed using the digits 1, 2, 3, 4 when repetition is allowed is
1) 84 2) 148 3) 180 4) 1440

PERMUTATIONS OF DISSIMILAR THINGS

6. The letters of the word 'HEXAGON' are arranged in all possible ways. If the order of the vowels is not to be considered then the number of possible arrangements is
1) 1680 2) 840 3) 420 4) 8!/2!

7. The letters of the word 'MADHURI' are arranged in all possible ways. The number of arrangements in which there are 2 letters between R and D is
1) 360 2) 480 3) 960 4) 720

8. The number of ways of arranging 8 Eamcet Question papers so that best and worst never come together is
1) 30240 2) 21600 3) 5040 4) 4320

9. The number of even numbers between 200 and 20,000 can be formed with the digits 0, 1, 2, 3, 4 when repetitions are not allowed is
1) 144 2) 120 3) 99 4) 96

10. The number of ways to arrange the letters of the word 'GARDEN' with vowels in alphabetical order is
1) 360 2) 240 3) 120 4) 480

SUM OF THE NUMBERS

11. The total number of seven-digit numbers such that the sum of whose digits is even is
1) 9×10^6 2) 45×10^5 3) 81×10^5 4) 9×10^5

12. The sum of all the numbers that can be formed by taking all digits 2, 3, 4, 4, 5 only is
1) 2399976 2) 2339976
3) 2333976 4) 2399376

PROBLEMS BASED ON REPETITIONS & SIMILAR THINGS

13. The number of ways in which six '+' and four '-' signs can be arranged in a line such that no two '-' signs come together is
1) 35 2) 120 3) 720 4) 610

14. The number of ways in which the letters of the word "SUCCESSFUL" be arranged such that the 'S's and 'U's will come together is
1) $7!$ 2) $\frac{7!}{2!}$ 3) $\frac{7!}{2!\,2!}$ 4) $\frac{7!}{2!\,2!\,2!}$

15. The number of ways in which the letters of the word "SUCCESSFUL" be arranged such that No two 'S's will come together is
1) $\frac{7!}{2!\,2!}$ 2) $\frac{7!}{2!\,2!}\cdot {}^{8}P_{3}$ 3) ${}^{8}P_{3}$ 4) $\frac{7!}{2!\,2!}\frac{{}^{8}P_{3}}{3!}$

16. The number of 6 digited numbers less than 4,00,000 can be formed by using the digits 1, 2, 3, 3, 3, 4 is
1) 150 2) 100 3) 50 4) 200

17. A library has 6 copies of one book 4 copies of each of two books, 6 copies of each of three books and single copies of 8 books. The number of arrangements of all the books in a linear shelf is
1) $\frac{40!}{(2!)^4\,(3!)^6}$ 2) $\frac{40!}{6!\,.\,(4!)^2\,(6!)^3}$
3) $\frac{40!}{6!\,.\,4!\,.\,6!}$ 4) $\frac{40!}{4!\,.\,(4!)^3\,.\,6!}$

18. The number of different numbers each of six digits that can be formed by using the digits of the numbers 2,2,3,3,9,9 is

1) 90 2) 100 3) 600 4) 120

DEARRANGEMENT PROBLEMS

19. There are four balls of different colours and four boxes of colours, same as those of the balls. The number of ways in which the balls, one each in a box, could be placed such that every ball does not go to a box of its own colour is

1) 9 2) 13 3) 8 4) 4

20. There are seven greeting cards, each of a different colour and seven envelopes of the same seven colours. The number of ways in which the cards can be put in the envelopes so that exactly four of the cards go into the envelopes of the right colours is

1) 2 x $^7C_3$ 2) $^7C_3$

3) 3! x $^4C_3$ 4) 3! x $^7C_3$ x $^4C_3$

CIRCULAR PERMUTATIONS

21. The number of ways in which 7 Indians and 6 Pakistanies sit around a round table so that no two Indians are together is

1) $(7!)^2$ 2) $(6!)^2$ 3) 6! 7! 4) zero

22. Number of ways in which 7 seats around a table can be occupied by 15 persons is

1) $^{15}P_7$ 2) $^{15}C_7/7$ 3) $^{15}P_7/7$ 4) 14!

OTHER MODELS

23. A binary sequence is an array of 0's and 1's. The number of n-digit binary sequences which contain even number of 0's is [EAM 2009]

1) 2^{n-1} 2) 2^n-1 3) $2^{n-1}-1$ 4) 2^n

24. The last digit of $(1!+2!+3!+......+2005!)^{500}$ is

1) 9 2) 2 3) 7 4) 1

25. The number of positive terms in the sequence $\{x_n\}$ where $x_n = \dfrac{195}{4(n!)} - \dfrac{^{n+3}P_3}{(n+1)!}$ are

1) 10 2) 5 3) 6 4) 4

LEVEL-II (H.W.) - KEY

1) 1	2) 3	3) 1	4) 2	5) 2	6) 2	7) 3
8) 1	9) 3	10) 1	11) 2	12) 1	13) 1	14) 2
15) 4	16) 2	17) 2	18) 1	19) 1	20) 1	21) 4
22) 3	23) 1	24) 4	25) 4			

LEVEL-II (H.W.) - HINTS

1. $^4P_3$
2. $^nP_2$
3. Required numbers = $3\times5\times5\times5$
4. The number of n-digit distinct numbers that can be formed using digits 2, 5, 7, is 3^n So $3^n \ge 900$

   $3^{n-2}\ge 100$; $n-2\ge5$; $n\ge7$
5. $4+4^2+4^3+1\times4^3=148$
6. $^7P_4$
7. $2\times{}^5P_2\times4!$
8. $8!-2!\times7!$
9. 21 + 60 + 18 (sum of the 3 digtis even + 4 digits even + 5 digits even)
10. $^6P_{6-2}={}^6P_4$
11. Suppose $a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7$ represents a seven digit number. Then a_1 taken the value 1, 2, ... 9 and a_2, a_3 ... a_7 all take values 0, 1, 2, ... 9. If we keep a_1, a_2, ... a_6 fixed then the sum ..... $a_1+a_2+....+a_6$ is either even or odd. Since a_7 taken 10 values 0, 1, 2 ... 9. Five of the numbers so formed will be even and 5 odd.

    $\therefore$ Required numbers =
    9.10.10.10.10.10.5=45x10^5.
12. 12 (2+3+5+4+4) (1 1 1 1 1) = 2399976
13. 6 '+'s arranged in $\dfrac{6!}{6!}$ ways and among them in 7 gaps 4 '-'s arranged in $\dfrac{^7P_4}{4!}$ ways

    $\therefore$ required ways = $\dfrac{6!}{6!}\times\dfrac{^7P_4}{4!}=35$
14. $\dfrac{SSS}{1}+\dfrac{UU}{1}+\dfrac{CCEFL}{5}$ = 7 objects

    $\therefore$ Ans = $\dfrac{7!}{2!}$

15. First arrange U C C E F U L in $\frac{7!}{2!\,2!}$ ways among these, there are 8 gaps arrange 3S's. In $\frac{^8P_3}{3!}$ ways

16. $\frac{5\times 5!}{3!}=100$

17. Total number of copies is
$(6\times4)+(6\times4)+(1\times8)=40$;
Required ways $\frac{\lfloor 40}{\lfloor 6(\lfloor 4)^2(\lfloor 6)^3}$

18. $\frac{6!}{2!.2!.2!}$

19. $4!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right)$

20. $^7P_3\left(\frac{1}{2!}-\frac{1}{3!}\right)=70=2.7_{C_3}$

21. In between 6 Pakistanies we have 6 gaps on a circular table, so 7 Indians cannot be arranged in 6 gaps.

22. $^nP_r/r$

23. The number of n digit binary sequences $=2^n$
The number of n digit binary sequences which contain even number of 0's is $\frac{2^n}{2}=2^{n-1}$

24. Let N = 1! + 2! + 3! + ... + 2005! = (1! + 2!+3!+4!) + (5! + 6! + ... + 2005!) = 33 + an integer having '0' in its unit place = an integer having 3 in its unit's place.
Hence N^{500} is an integer having '1' in its units place.

25. If n = 1, 2, 3, 4 then $x_n > 0$
Number of positive terms is 4.

LEVEL-III

1. The range of the function $f(x)={}^{7-x}P_{x-3}$ is
1) {1, 2, 3} 2) {1, 2, 3, 4}
3) {1, 2, 3, 4, 5} 4) {1, 2, 3, 4, 5, 6}

2. When n is a positive integer, then $(n^2)!$ is
1) divisible by n^2, but not by n!
2) divisible by n!, but not by $(n^2)!$
3) divisible by $(n^2)!$, but not by $(n!)^n$
4) divisible by $(n!)^n$

3. The number of distinct rational numbers p/q, where p,q $\in$ {1, 2, 3, 4, 5, 6} is
1) 23 2) 32 3) 36 4) 63

4. If m = number of distinct rational numbers p/q $\in$ (0, 1) such that p, q $\in$ {1, 2, 3, 4, 5} and n = number of onto mappings from {1, 2, 3} onto {1, 2}, then m - n is
1) 1 2) -1 3) 0 4) 3

5. How many numbers lying between 100 and 10,000 can be formed with the digits 1,2,3,4,5 if atleast one digit in the same numbers are repeated
1) $750-{}^5P_3-{}^5P_4$ 2) $750-{}^5P_3+{}^5P_4$
3) $750+{}^5P_3-{}^5P_4$ 4) $750+{}^5P_3+{}^5P_4$

6. If all permutations of the letters of the word "AGAIN" are arranged as in dictionary, then fiftieth word is
1) NAAGI 2) NAGAI 3) NAAIG 4) NAIAG

7. The number of permutaions of the letters of the word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears, are :
1) 166674 2) 169194 3) 166680 4) 181434

8. The tens' digit of $\lfloor 1+\lfloor 2+\lfloor 3+....+\lfloor 29$ is
1) 1 2) 2 3) 3 4) 4

9. 5 students (A,B,C,D,E) are to be arranged in a row so, that A occupies the 2nd position and B is always adjacent to 'c', the number of such arrangements is
1) 6 2) 4 3) 8 4) 2

10. No of different matrices that can be formed with elements 0,1,2, or 3 each matrix having 4 elements is
1) 3×2^4 2) 2×4^4 3) 3×4^4 4) 4^4

11. Sum of four digit numbers formed with 2,3,4,5, using each digit any number of times is
1) $4\times1111\times64$ 2) $14\times1111\times64$
3) $14\times1111\times16$ 4) $4\times1111\times16$

12. Total number of numbers that are less than 3×10^8 and can be formed using the digits 1, 2, 3 is equal to
1) $\frac{1}{2}(3^9+4.3^8)$ 2) $\frac{1}{2}(3^9-3)$
3) $\frac{1}{2}(7.3^8-3)$ 4) $\frac{1}{2}(3^9-3+3^8)$

13. The number of numbers between 1 and 10^{10} which contain the digit 1 is :
1) $10^{10} - 9^{10} + 1$ 2) $10^{10} - 9^{10} - 1$
3) $10^{10} - 8^{10} + 1$ 4) $10^{10} - 9^{10} + 2$

14. No of symmetric matrices of order 3×3 by using the elements of the set $A=\{-3,-2,-1,0,1,2,3\}$ is
1) 7^9 2) 7^6 3) 7^3 4) 7^9-7^3

15. In the above problem No.of symmetric (or) skew symmetric matrix is
1) 7^6+7^3-1 2) 7^9+7^6 3) 7^9-7^6 4) 7^9-7^6-1

16. In a particular programming language a valid variable name can consist of a sequence of one to five alphanumeric characters A,B,C,D,......,Z, 0,1,2,.....,9 beginning with a letter. The number of valid variable name is
1) $\frac{26}{25}\left(26^5-1\right)$ 2) $\frac{26}{25}\left(26^5+1\right)$
3) $\frac{26}{35}\left(36^5+1\right)$ 4) $\frac{26}{35}\left(36^5-1\right)$

17. If N is the sum of all digits used in writing all the numbers from 1 to 1000, then the last digit of N is
1) 1 2) 2 3) 3 4) 4

18. Consider 10 digits numbers of the form $M=\sum_{r=0}^{9}a_r10^r$,where $a_r, r=0,1,2,.........9$ are distinct digits, $a_9\neq 0$, which are divisible by 99999. If N is the number of such numbers, then the last digit of N is
1) 6 2) 7 3) 8 4) 9

19. The number of arrangements of the letters of the word FORTUNE such that the order of vowels is unaltered is x, the order of consonents is unaltered is y, the order of vowels and consonents is unaltered is z, then $\frac{x+y}{z}$ equals
1) 10 2) 20 3) 30 4) 40

20. A variable name with atmost two characters in certain computer language must be either a alphabet or a alphabet followed by using the digit. Total number of different variable names that can exist in that language is equal to:
1) 280 2) 290 3) 286 4) 296

LEVEL - III - KEY

1) 1	2) 4	3) 1	4) 4	5) 1	6) 3	7) 2
8) 1	9) 3	10) 3	11) 2	12) 3	13) 2	14) 2
15) 1	16) 4	17) 1	18) 1	19) 3	20) 3	

LEVEL - III - HINTS

1. ${}^{7-x}P_{x-3}$ is defined for x if x - 3 $\geq$ 0 and 7-x$\geq$x-3 $\Rightarrow$ x $\leq$ 5. The domain is {3, 4, 5}
$f(3)={}^4P_0=1,\ f(4)={}^3P_1=3,$
$f(5)={}^2P_2=2$. The range is {1, 2, 3}

2. Consider n identical objects of first kind, n identical objects of second kind, etc and n identical objects of n^{th} kind. There are in all n^2 objects.
The number of their arrangements is $\frac{\left(n^2\right)!}{\left(n!\right)^n}$ is an integer. $\left(n^2\right)!$ is divisible by $\left(n!\right)^n$

3. The given 6 digits can occupy the numerator and denominator places in ${}^6P_2=30$ ways. But out of them $\frac{1}{2},\frac{2}{4},\frac{3}{6}$ represent same number. $\frac{2}{1},\frac{4}{2},\frac{6}{3}$ represent same number. $\frac{2}{3},\frac{4}{6}$ represent same number. Similarly $\frac{3}{2},\frac{6}{4}$ represent same number. $\frac{1}{3},\frac{2}{6}\ \&\ \frac{3}{1},\frac{6}{2}$ represent same number. In all above cases we have to consider only one case each.
$\therefore$ No. of rational numbers
= 30 - (2 + 2+ 1 + 1+ 1 + 1)+1
= 22 + 1 ($\because$ '1' is a rational no) = 23

4. $m={}^5C_2-1=10-1=9;\left(\because=\frac{1}{2}\ \textit{is identical to}\ \frac{2}{4}\right)$
$n=2^3-2=6$

5. Total number of ways = $5^3+5^4=750$
atleast one repeated = Total number of ways - The number of ways without repetition.

6. Find the tens digit of 1! + 2! + 3! + 4! + 5! + 6! + 7! + 8! + 9!

7. Requried number $=\frac{9!}{2!}-\left(7!+7!+\frac{7!}{2}\right)+3\times5!-3!$

$=169194.$

8. Add $1!+2!+3!+4!+5!+.........+9!$ and find the digit in 10's place, it is 1 (From 10! onwards the units and 10's places is zero only).

9. 4 x 2! = 8

A is in second position and B and C can be together in 4 ways and remaining 2 places is filled in 2 ways.

10. The possible sizes of matrices haveing 4 elements are $1\times4,\ 4\times1,\ 2\times2;\ (i.e\ 3\ ways)$

4 placess in each matrix can be filled in 4^4 ways

$\therefore$ No. of different matrices $=3\times4^4$

11. $Sum=(2+3+4+5)\times4^3\times(1111)$

12. Formed number can be atmost of nine digits. Total number of such numbers

$=3+3^2+3^3+......+3^8+2.3^8$

13. No.of numbers = Total Possible numbers by using all digits - Total possible numbers without '1' -1

$=10^{10}-9^{10}-1$

14. Diagonal elements are arranged in 7^3 ways and all the remaing elemts are arranged in $7^3\times1^3$ ways

$\therefore$ Total arrangement $=7^3\times7^3\times1^3=7^6$

15. Required number = No of symmetic + No of skew symmetic-1 $=7^6+7^3-1$

16. The first entry is a letter. The other entries are letters or digits.

The desired number is $\sum_{r=0}^{4}26.36^r=\frac{26}{35}\left(36^5-1\right).$

17. We write all the numbers from 0 to 999 as 3 digits numbers: 000, 001, 002, ...998, 999

Writing them in the reverse order, 999, 998, 99 ....001, 000

Adding the above 2000 numbers, we get 999 999 .... 999 999

$\therefore$ The sum of the digits of the 1000 number from 0 to 999 is $\frac{27000}{2}=13500$

Now delete 0 and add 1000 to the list of 1000 numbers. $N=13501$.

18. $M=\sum_{r=0}^{9}a_r10^r=\left(a_9a_8a_7a_6a_5\right)10^5+a_4a_3a_2a_1a_0$

$=\left(10^5-1\right)a_9a_8a_7a_6a_5+a_9a_8a_7a_6a_5+a_4a_3a_2a_1a_0$

$\therefore$ 99999 divides the sum $a_9a_8a_7a_6a_5+a_4a_3a_2a_1a_0$

$\therefore a_9+a_4=a_8+a_3=a_7+a_2=a_6+a_1=a_5+a_0=9$

The pairs $(a_9,a_4),(a_8,a_3),(a_7,a_2),(a_6,a_1),(a_5,a_0)$ can be chosen in 9,8,6,4,2 was respectively.

$\therefore N=9\times8\times6\times4\times2=3456$.

19. $x=\frac{7!}{3!},\ y=\frac{7!}{4!},\ z=\frac{7!}{3!4!}\quad\Rightarrow x+y=30z$

20. Total variables if only alphabet is used = 26

Total variables if alphabets and digits both are used $=26.10\Rightarrow$ Total variables $=26(1+10)=286.$

LEVEL-IV

STATEMENT TYPE

1. I : The number of five digit numbers that canbe formed with 0, 1, 2, 3, 5, which are divisible by 5 is 18.

II : The sum of all four digit numbers that can be formed with 1, 2, 3, 4 so that no digit being repeated in any number is 66660.

which of the above statements is true?

1) only I 2) only II

3) Both I and II 4) Neither I nor II

2. I : The number of ways that 5 prizes be distributed among 4 boys while each boy is eligible for any number of prizes is 4^5

II : The letters of the word "TOSS" are permuted in all possible ways and the words thus formed are arranged as in a dictionary, the rank of the word "TOSS" is 10

Which of the above statements is correct

1) only I 2) only II

3) both I and II 4) Neither I nor II

3. I. The no.of ways in which 4 letters can be posted in 5 letter boxes is 4^5 ways.

II. If words are formed by taking only 4 at a time out of the letters of the word PHYSICS, then the number of words in which Y occur is 288. Which of the above statements is correct

1) only I 2) only II

3) both I and II 4) Neither I nor II

4. **I : The no.of permutations of n different things taken any no.of things at a timeis $^{n}P_{n}$ (repetition not allowed)**
II. The no.of permutations of n different things taken any no.of things at a time is
$^{n}P_{1}+^{n}P_{2}+^{n}P_{3}+.....+^{n}P_{n}$
(repetition not allowed)
Which of the above statements is correct?
1) only I 2) only II
3) Both I and II 4) Neither I nor II

5. **If the letters of the word PRISON are permuted in all possible ways and the words thus formed are arranged in dictionary order then**
I: The rank of PRISON is 438
II: The rank of SIPRON is 618
1) only I is ture
2) only II is true
3) both I and II are true
4) neither I nor II true

6. **I: The number of ways of arranging 4 boys and 3 girls arround a circle so that all the girls sit together is 144.**
II: The number of ways of arranging 6 red roses and 3 yellow roses of different sizes into a garlend so that no to yellow roses come together is 7200
1) only I is ture
2) only II is true
3) both I and II are true
4) neither I nor II true

ASCENDING & DESCENDING ORDERS

7. **The letters of the following words are arranged and words thus formed are kept as in dictionary. Then arrange the following in the descending order of there ranks**
A : SITA B : RAMU C : TEA D : BUT
1) DCBA 2) ABDC
3) ABCD 4) BACD

8. **If a, b, c are the number of 4 digited numbers that can be formed using the digits 2, 4, 5, 7, 8 that are divisible by 3, 4, 5 respectively then the ascending order of a, b, c is**
1) a, b, c 2) b, c, a
3) c, b, a 4) c, a, b

9. **If a, b, c are the number of functions, injections, constant functions from a set containing 3 elements into a set containing 6 elements respectively then the ascending order of a, b, c is**
1) a,b,c 2) b,c,a 3) c,b,a 4) c,a,b

10. **Of the 4 letter words formed by using the letters of the word EQUATION, a, b, c are respectively the number of words begin with an vowel, begin and end with vowels, end with a consonent then the ascending order of a, b, c is**
1) a,b,c 2) b,c,a 3) c,b,a 4) a,c,b

ASSERTION & REASON

Choose the correct Answer
1) A is true, R is true and R is the correct explanation for A
2) A is true, R is false and R is the not the correct explanation for A
3) A is true, R is false
4) A is false, R is true.

11. **Assertion (A) :** The number of ways of arranging 6 boys and 5 girls alternately at circular table is 0
Reason (R) : To arrange boys and girls alternately at a circular table, they should be equal in number.

12. **Assertion(A):** The number of quadratic expressions with the coefficients drawn from the set $\{0,1,2,3\}$ is only 48 but not 64.
Reason(R): The coefficient of x^2 in the quadratic expression ax^2+bx+c can not be '0'.

13. **Assertion:** There are three doors to a room. The number of ways in which a student can enter the room and leave it by a different door is 6.
Reason: If an operation can be performed in m ways and another operation can be performed in n ways then the two operations in succession can be performed in mn ways.

14. **Assertion:** The sum of all 4 digited numbers that can be formed using the digits 1,2,4,5,6 without repetition is 479952
Reason: Sum of all r-digit numbers formed by taking the given n digits (without zero) is (sum of all the n digits) $\times\, ^{(n-1)}P_{r-1}\times(111....r\text{ times})$.

MATCHING TYPE

15. Match the following:

1. If ${}^{n}P_5 : {}^{n}P_3 = 2:1$ then n = a) 16
2. If ${}^{12}P_6 + 6.{}^{12}P_5 = {}^{13}P_n$ then n = b) 8
3. If ${}^{n}P_4 = 1680$ then n = c) 6
4. If ${}^{n}P_5 = 12\ {}^{n}P_4$ then n = d) 5

1) a,b,c,d 2) b,c,a,d 3) c,d,b,a 4) d,c,b,a

16. Match the following:

Given Word	**Rank**
I. REMAST	a) 616
II. MASTER	b) 391
III. LUBER	c) 257
IV. ZENITH	d) 67

1) a,b,c,d 2) b,c,a,d
3) c,d,b,a 4) b,c,d,a

LEVEL - IV - KEY

1) 2	2) 3	3) 2	4) 2
5) 3	6) 3	7) 3	8) 3
9) 3	10) 2	11) 1	12) 1
13) 1	14) 1	15) 4	16) 4

LEVEL - IV - HINTS

1. I) If 0 is in units place (divisible by 5) in ${}^{4}P_3$=24.
If 5 is in units place (divisible by 5) in ${}^{4}P_3 - {}^{3}P_2 = 18$.
Total : 24 + 18 = 42
II) ${}^{3}P_3(1+2+3+4)(1111) = 66660$

2. From synopsis I : 4^5;
II : O S S T ; TOSS
$\frac{3}{2!}.3!+0.2!+0.1!+1 = 10$

3. I : 1 letter is posted in 5 ways $\Rightarrow 5^4$
II: The given word is PHYSICS
1) All different along with Y = ${}^{5}C_3$ x4! = 240
2) Two same along with Y $= {}^{4}C_1$ x $\frac{4!}{2!} = 48$
Total = 288

4. From synopsis II statement is true

5. I: $3\times5!+3\times4!+0\times3!+2\times2!+1\times1!+1 = 438$
II: $5\times5!+0\times4!+2\times3!+2\times2!+1\times1!+1 = 618$

6. I: $4!\times3! = 144$
II: $\frac{5!\times6_{P_3}}{2} = 7200$

7. SITA - AIST = 2.3! + 1.2! + 1.1! + 1 = 16
RAMU - AMRU = 2.3! + 0.2! + 0.1! + 1 = 13
TEA - AET = 2.2! + 1.1! + 1 = 6
BUT - BTU = 0.2! + 1.1! + 1 = 2
Desending order ABCD

8. $(4,5,7,8),(2,4,7,8),(2,4,5,7)$
$a = 4! + 4! + 4! = 96$
$b = 6\times3_{p_2} = 36$
$c = 1\times4_{p_3} = 24$

9. a = $6^3 = 216$
b = $6_{P_3} = 120$
c = 6

10. a = $5_{P_1}\times7_{P_3} = 1050$
b = $5_{P_2}\times6_{P_2} = 600$
c = $3_{P_1}\times7_{P_3} = 630$

11. Reason is correct explanation of the statement

12. Reason is correct explanation of Assertion.

13. A = $3\times2 = 6$

14. A = $4_{P_3}\times(1+2+4+5+6)\times1111 = 279952$

15. Simplify (Conceptual)

16. "REMAST" rank =
$3\times5!+1\times4!+1\times3!+0\times2!+0\times1!+1 = 391$
"MASTER" rank =
$2\times5!+0\times4!+2\times3!+2\times2!+0\times1!+1 = 257$
"LUBER" rank =
$2\times4!+3\times3!+0\times2!+0\times1!+1 = 67$
"ZENITH" rank =
$5\times5!+0\times4!+2\times3!+1\times2!+1\times1!+1 = 616$